Soft vs Rigid Gripper Benchmark for Cloth Grasping
A Force-Depth-Tilt Characterization of Parallel-Jaw and Finray Geometries

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Abstract

Robotic grasping of thin, deformable fabric remains an unsolved problem in industrial automation. Two competing gripper geometries are routinely proposed: rigid parallel-jaw fingers, which offer precise force control but a narrow operational tolerance, and compliant finray-effect fingers, which trade force precision for a wider operational envelope. This thesis presents an experimental benchmark of six gripper variants: two parallel-jaw configurations (rigid stock, and stock plus a Shore-90A thermoplastic polyurethane pad) and four finray configurations (18 mm and 26 mm finger thicknesses, a 5° inclined-seat fixture, and a second print of the 18 mm finger), all mounted on a Franka FR3 collaborative manipulator. Each variant was characterised across a two-dimensional input space of press depth (the distance the gripper fingertip is driven below the detected cloth surface) and tilt angle (the gripper orientation about world x and y axes relative to the surface normal). Stationary contact force in the gripper z-axis was sampled during each grasp and used as the primary outcome variable, with binary grasp success as a secondary outcome.

The benchmark produced 845 analysed trials across the six variants. The rigid parallel-jaw variant exhibits a narrow 1.5 mm working depth range bounded above by a -60 N safety reflex and below by no-contact; the TPU-padded variant widens this to 2.5 mm. The finray variants, by contrast, remain successful across the full 10 mm calibrated analysis window (working range $\geq$ 10 mm) with a soft force plateau past the buckling knee (knee peak forces of 31–53 N across the finray variants).

Beyond this expected compliance advantage, two depth-dependent tilt effects emerged. First, finray contact force is strongly tilt-sensitive at the no-contact boundary but robust to tilt in the soft plateau: the signed force asymmetry between opposite 2.5° tilts peaks at 5.4 N at boundary depths and falls below 0.9 N at every depth

beyond $+4\,\mathrm{mm}$. Second, the finray gripper exhibits a direction-dependent handedness under tilt about its asymmetric (y) axis at boundary depths (a $\pm 2.5°$ inversion changes the contact force by a factor of 2.36 at $-1.0\,\mathrm{mm}$), but this handedness vanishes at safe operating depths. This is proposed as an emergent characterisation of the finray geometry: the deeply pressed finger structure wraps the contact surface symmetrically regardless of tilt sign, while at lighter contact the asymmetric ridge profile dominates the force response.

The implication for cloth-handling system design is that compliant grippers achieve directional robustness only when pressed past a depth threshold; control policies that operate near the no-contact boundary for delicate fabric handling must explicitly model the gripper's directional asymmetry.

Abrégé

La préhension robotique de tissus minces et déformables demeure un problème non résolu en automatisation industrielle. Deux géométries de préhenseur concurrentes sont couramment proposées : les doigts rigides à mâchoires parallèles, qui offrent un contrôle de force précis mais une tolérance opérationnelle étroite, et les doigts souples à effet finray, qui échangent la précision de force contre une enveloppe opérationnelle plus large. Cette thèse présente un banc d'essai expérimental de six variantes de préhenseur : deux configurations à mâchoires parallèles (rigide d'origine, et d'origine avec un coussinet en polyuréthane thermoplastique de dureté Shore 90A) et quatre configurations finray (doigts de 18 mm et de 26 mm d'épaisseur, une fixation inclinée à 5°, et une seconde impression du doigt de 18 mm), toutes montées sur un manipulateur collaboratif Franka FR3. Chaque variante a été caractérisée sur un espace d'entrée bidimensionnel composé de la profondeur d'appui (la distance à laquelle le bout du doigt est enfoncé sous la surface détectée du tissu) et de l'angle d'inclinaison (l'orientation du préhenseur autour des axes x et y du repère monde par rapport à la normale de la surface). La force de contact stationnaire selon l'axe z du préhenseur a été échantillonnée pendant chaque prise et a servi de variable de résultat principale, la réussite binaire de la prise servant de résultat secondaire.

Le banc d'essai a produit 845 essais analysés sur les six variantes. Le préhenseur rigide à mâchoires parallèles présente une plage de profondeur utile étroite de 1,5 mm, bornée supérieurement par un réflexe de sécurité à -60 N et inférieurement par l'absence de contact ; la variante avec coussinet TPU élargit cette plage à 2,5 mm. Les variantes finray, en revanche, restent efficaces sur toute la fenêtre d'analyse calibrée de 10 mm (plage utile de 10 mm ou plus), avec un plateau de force souple au-delà du coude de flambement (forces de pic au coude de 31 à 53 N selon la variante).

Au-delà de cet avantage attendu de la souplesse, deux effets d'inclinaison dépendants de la profondeur ont émergé. Premièrement, la force de contact du finray est fortement sensible à l'inclinaison près de la limite de non-contact, mais robuste à l'inclinaison dans le plateau souple : l'asymétrie signée de force entre deux inclinaisons opposées de 2,5° culmine à 5,4 N aux profondeurs limites et tombe sous 0,9 N à toute profondeur au-delà de +4 mm. Deuxièmement, le préhenseur finray présente une réponse directionnelle (« chiralité ») sous inclinaison autour de son axe asymétrique (y) aux profondeurs limites (une inversion de $\pm$2,5° modifie la force de contact d'un facteur de 2,36 à $-$1,0 mm), mais cette chiralité disparaît aux profondeurs d'opération sûres. Ceci est proposé comme une caractérisation émergente de la géométrie finray : le doigt profondément appuyé épouse la surface de contact de manière symétrique quel que soit le signe de l'inclinaison, tandis qu'à contact plus léger, le profil asymétrique de l'arête domine la réponse en force.

L'implication pour la conception de systèmes de manipulation de tissus est que les préhenseurs souples n'atteignent une robustesse directionnelle qu'au-delà d'un seuil de profondeur d'appui ; les politiques de commande qui opèrent près de la limite de non-contact pour la manipulation délicate de tissus doivent explicitement modéliser l'asymétrie directionnelle du préhenseur.

[TODO: Faire vérifier cette traduction par un collègue francophone avant le dépôt final (exigence de qualité, non de conformité : GPS exige seulement la présence d'un abrégé en français).]

Contribution to Original Knowledge

This thesis contributes the first controlled, parametric characterisation of how compliant finray-effect fingers and rigid parallel-jaw grippers grasp cloth as a function of press depth and gripper tilt, holding the robot, control policy, cloth specimen, and environment fixed and varying only the end-effector. Prior work on finray-family fingers has modelled or measured *lateral* grasping contact; no published model or benchmark, to the author's knowledge, characterises the axial tip-compression response: the rise–peak–valley–recovery force profile identified here, shared across four finray variants to within 1% normalised error. The principal empirical contribution is the demonstration that finray tilt robustness is *depth-gated*: strongly tilt-sensitive near the no-contact boundary, but robust to tilt once pressed past the buckling threshold, with a direction-dependent asymmetry about the finger's geometrically asymmetric axis that likewise vanishes in the safe operating regime. This depth-dependent tilt robustness had not been quantitatively characterised in prior literature. Secondary contributions include a reproducible benchmark protocol and its open-source implementation, a documented per-variant tool-centre-point calibration procedure, and a geometric verification framework disentangling control-frame positioning error from physical gripper compliance under load.

Contribution of Authors

This thesis is the original work of the author, Shiyao Ni, under the supervision of Prof. Audrey Sedal. The author carried out all experimental design, hardware integration, software development, data collection, analysis, and writing.

The Franka FR3 manipulator was maintained by the MACRObotics Lab; the `fr3_pose_controller` ROS 2 control stack was developed and maintained by the McGill Applied Dynamics Lab. The 3D-printed finray fingers were fabricated using lab resources at the MACRObotics Lab. No portion of this thesis has been previously published.

This is a traditional (monograph-style) thesis; no chapter derives from a co-authored manuscript, and no collaborator contributed content requiring declaration under McGill GPS guidelines beyond the infrastructure acknowledgements above.

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I thank my supervisor, Prof. Audrey Sedal, for her guidance throughout this work, and the members of the MACRObotics Lab at McGill University for their support in the laboratory. I am grateful to the McGill Applied Dynamics Lab for developing and maintaining the `fr3_pose_controller` ROS 2 control stack on which every experiment in this thesis depends, and to the broader ROS 2 and Franka robotics communities whose open-source tooling made this work possible.

Finally, I thank my family for their patience and encouragement.

[TODO: Add funding sources (scholarships, grants, or stipend support), if any, per McGill GPS acknowledgement conventions.]

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List of Abbreviations

API	Application Programming Interface
CAD	Computer-Aided Design
DOF	Degree(s) of Freedom
FCI	Franka Control Interface
FR3	Franka Research 3 (robot model)
GT	Ground Truth (calibrated surface height record)
NMSE	Normalised Mean-Squared Error
PEI	Polyetherimide (3D-printer build-plate surface)
RMSE	Root-Mean-Square Error
RGB-D	Red–Green–Blue plus Depth (camera modality)
ROS 2	Robot Operating System 2
RPY	Roll–Pitch–Yaw (orientation angles)
TCP	Tool Centre Point
TPU	Thermoplastic Polyurethane
URDF	Unified Robot Description Format

Notation

Frames and poses

$\mathcal{W}$, $\mathcal{F}$, $\mathcal{T}$	World (robot base), gripper flange, and gripper tip frames
$\mathbf{p}$	Position vector in $\mathbb{R}^3$ (boldface); $\mathbf{p}_{\mathcal{F}}$, $\mathbf{p}_{\mathcal{T}}$ denote the flange and tip positions expressed in $\mathcal{W}$
R, R_0	Orientation of the gripper as a rotation matrix, $R \in SO(3)$; $R_0 = R_x(180°)$ is the nominal down-pointing orientation
$R_x(\phi)$, $R_y(\phi)$	Elementary rotation matrix for a rotation by angle ϕ about the fixed world x or y axis
$\hat{\mathbf{z}}$	Unit vector along $+z$
z_{TCP}	Per-variant tool-centre-point offset from flange to tip along the flange's local z axis (mm); $z_{\text{TCP}}^{(v)}$ denotes the calibrated value for variant v
x_0, y_0	Fixed probe/target position in the world xy plane (mm)

Design parameters (fixed per variant)

t	Finray finger blade thickness (mm) (Fig. 3.1)
α	Baseline seat-mount tilt angle, about the gripper's local y axis, introduced by the 3D-printed inclined seat (°) (Fig. 3.1)

Runtime and trial variables

θ_x, θ_y	Commanded robot-arm tilt angles about the world x / y axes (which coincide with the gripper's local x / y axis lines at R_0), applied as $R_{\text{tilt}} = R_y(\theta_y)\, R_x(\theta_x)$ (extrinsic) (°); distinct from, and never summed with, the seat angle α (Fig. 3.3)
d	Press depth: distance the gripper fingertip is driven below the detected cloth/table surface (mm)
d_{cal}	Calibrated press depth: $d - d_{\text{first success}}$, zeroed at each series' first majority-success cell (mm)
z_{surf}	Detected cloth/table surface height at the tip's xy location (mm)
a	Tilt axis identifier (x or y)
$b(d, a)$	Signed tilt bias: $b(d, a) = \lvert F_z^{+2.5°}(d, a)\rvert - \lvert F_z^{-2.5°}(d, a)\rvert$ (N)

Force measurement

$\mathbf{F}_{\text{ext}}$	Franka external wrench force vector at the end-effector flange (N); F_x, F_y, F_z denote its components
F_z^{bias}	Free-air F_z bias captured at session start (N)
$F_z^{(i)}$	The i-th stationary F_z sample within a trial (N)
N	Number of stationary force samples pooled in a (variant, depth, tilt) cell
$\bar{F}_z$, σ_{F_z}	Per-cell mean and standard deviation of F_z over pooled trials
$\sigma_{z,\text{tcp}}$	Across-tap standard deviation of the touch-off z measurement during TCP calibration (mm; distinct from σ_{F_z} above)

Derived quantities

L_{eff}	Effective finger lever-arm length inferred from the mount-tilt/depth-shift equivalence (mm)

Preface

This work grew out of a practical frustration: in the laboratory, as in much of the published literature, robotic cloth-manipulation demonstrations routinely begin with a human placing the fabric into the robot's gripper. The step that most limits autonomy, acquiring the first grasp of a thin, flat cloth lying on a surface, is the step most often skipped. Rather than approaching that gap with more perception or learning, this thesis asks a hardware question: how much of the problem dissolves if the gripper itself is compliant, and what, precisely, is traded away in exchange?

All chapters of this thesis are original, previously unpublished work by the author. Collaborative and infrastructural contributions are declared in the Contribution of Authors section.

1

Introduction

1.1 Motivation

Fabric-handling automation is a long-standing open problem in industrial robotics, with applications spanning apparel manufacturing and textile logistics, laundry processing and folding, hospital bedding management, assistive robotics for elderly or disabled individuals, and the packaging of soft goods. Yet a striking fraction of published robot-manipulation demonstrations sidesteps the hardest step entirely: the object is handed to the robot's gripper by a human operator, and the demonstrated autonomy begins only after the grasp already exists.[1] For cloth specifically, acquiring the first grasp (picking a thin, flat fabric directly off a surface) remains the bottleneck that limits how autonomous these systems can truly be. Cloth is uniquely difficult to grasp robustly: it deforms continuously under contact, exhibits no fixed graspable feature, and has a near-zero bending stiffness that allows it to slip out of a poorly

[1]A prominent recent example: a widely circulated January 2024 video of the Tesla Optimus humanoid folding a shirt was subsequently acknowledged by its poster to be teleoperated, with the shirt pre-staged in a fixed setup [1, 2].

placed gripper. Conventional rigid-jaw grippers, designed for sturdy objects with well-defined geometry, perform poorly on cloth without elaborate tactile sensing or human-tuned approach trajectories.

A class of compliant grippers based on the finray effect, a passive bending mechanism inspired by fish-fin anatomy, has been proposed as a solution. Finray fingers passively wrap around the contact surface, increasing contact area and gripping force without requiring active force control. This compliance, however, comes at the cost of force-modelling difficulty: the finger geometry deforms under load, the contact point shifts, and tilt-induced asymmetries arise that have no analogue in the rigid-jaw case.

The central question of this thesis is: *to what extent does the finray's compliance trade single-axis force precision for multi-axis operational robustness?* Specifically, how does the finray's contact force respond to deviations in the two design degrees of freedom most likely to be uncontrolled in real deployment: press depth and gripper tilt angle?

1.2 Research Questions and Contributions

This thesis answers the following research questions through a controlled experimental benchmark on a Franka FR3 collaborative robot. The benchmark treats the gripper itself as the experimental variable, holding the robot, control policy, cloth, and environment identical across all conditions, and asks:

1. How does the contact force vary with press depth, for each finger design?
2. How much can the gripper be tilted before the grasp becomes unreliable?
3. Does tilt tolerance depend on the press depth?
4. What is the soft gripper's response under compression?

The contributions of this work are:

- A reproducible benchmark protocol for cloth-grasp force characterisation on a Franka FR3, with full source code published alongside this thesis.

- A two-dimensional depth–tilt experimental matrix sampling six gripper variants: two rigid parallel-jaw configurations and four finray prints spanning blade thickness, seat angle, and print instance.
- Empirical evidence of a depth-dependent tilt robustness transition in finray grippers, with quantitative characterisation of the transition depth.
- A documented calibration procedure for per-variant tool-centre-point (TCP) offset using differential force touch-off against a rigid reference.
- A geometric verification framework that disentangles control-frame positioning error from physical gripper compliance under load.

Scope. The benchmark deliberately isolates the *contact* phase of cloth grasping, the interval between first gripper–cloth contact and a completed lift, and characterises it over a continuous depth–tilt parameter space. Downstream task-level evaluation (folding, placing, and dragging under the same policy) is future work (Chapter 7). This narrowing is intentional: it is precisely because the contact phase is usually bundled inside end-to-end task pipelines that its mechanics have not previously been mapped as a function of gripper design.

1.3 Thesis Organisation

Chapter 2 reviews the literature on cloth grasping, compliant grippers, and force-controlled approach strategies. Chapter 3 describes the experimental hardware, the kinematic calibration procedure, and the data-acquisition pipeline. Chapter 4 presents the two experimental campaigns: the depth-sweep characterisation (Experiment 1) across all six gripper variants at zero tilt, the tilt-sweep characterisation (Experiment 2) at boundary and safe depths, and the planned inclined-surface pilot demonstration (Experiment 3). Chapter 5 reports the resulting force–depth–tilt characterisation. Chapter 6 synthesises the findings into a model of depth-dependent tilt robustness. Chapter 7 concludes and identifies directions for future work.

2

Background and Related Work

2.1 Cloth Grasping in Robotics

Robotic manipulation of cloth and other deformable fabric is difficult precisely because the object being manipulated lacks a fixed geometry: unlike a rigid part, a cloth's shape is itself part of the manipulation state, and it must typically be inferred from vision rather than assumed from a physical model. Early work in this space focused on perceiving that shape well enough to act on it. Kita et al. [3] fit a deformable mass-spring model of a held garment to stereo-camera silhouette observations, using the best-matching simulated deformation to estimate the position of an unheld part of the garment for a second manipulator to grasp. This was an early demonstration that a physics-informed shape prior, rather than raw image features, is what makes cloth state estimation tractable. Maitin-Shepard et al. [4] took a complementary, model-light approach: a multiple-view geometric-cue detector locates cloth corners directly from silhouettes, in a manner robust to texture and colour variation, letting a general-purpose PR2 manipulator pick and fold towels of varying material and appearance without a specialised end-effector. Miller et al. [5] generalised this line of work with a polygonal-approximation geometric model that tracks a garment's silhouette through an entire folding sequence, allowing one parameterised folding strategy to transfer across previously unseen towels, pants, and shirts. Li et al. [6] pushed the physical modelling further still, using a predictive thin-shell simulation fit to RGB-D observations to plan regrasp actions that unfold and flatten a crumpled garment by selecting the grasp points expected to maximise resulting flatness.

More recent work has moved from hand-designed geometric priors toward learned policies and richer sensing. Saxena and Shibata [7] trained convolutional classifiers to recognise garment type and locate hem/waistline grasp points directly from images, as the perception front-end of a robotic dressing-assistance pipeline. Wu et al. [8] showed that a model-free visual reinforcement-learning policy for cloth and rope manipulation can be learned an order of magnitude faster than a jointly-learned pick-and-place policy by conditioning only on the *placing* action and selecting picks that maximise the value of the resulting placement, entirely without human demonstrations. Closest in spirit to the force-sensing emphasis of this thesis, Tirumala et al. [9] mounted a tactile sensor on a Franka gripper fingertip and trained a classifier on the resulting tactile signal to determine how many layers of cloth had been grasped, using that signal to re-grasp until exactly one or two layers were singulated from a stack. This is direct evidence that certain cloth-grasp ambiguities cannot be resolved by vision alone. For a comprehensive treatment of the field beyond these representative examples, Longhini et al. [10] provide a recent survey spanning perception, dynamics modelling, benchmarking, and control specifically for textile manipulation.

2.2 Strategies for Acquiring the First Grasp

Because a flat cloth on a surface presents no protruding feature for a conventional pinch grasp, the literature has developed two broad families of workarounds: manipulating the *environment or cloth state* until a graspable feature exists, and replacing the *end-effector* with a mechanism that does not require one.

2.2.1 Environment-Assisted and State-Changing Actions

The first family keeps a standard gripper and instead spends actions creating a graspable feature (a fold, wrinkle, or free edge) before the pick. Ramisa et al. [11] detected and ranked wrinkles in depth images of crumpled laundry, directing a standard gripper to the most graspable bump on the cloth rather than to a flat region. Qian et al. [12] segmented cloth edges and corners (as distinct from wrinkles and folds) in depth images and coupled the segmentation to a sliding grasp policy that approaches along the estimated grasp direction. At the dynamic extreme, Ha and Song's FlingBot [13] showed that a high-velocity fling primitive can unfold a crumpled cloth in a few actions: environment interaction used not to create a pinchable feature but to reset the cloth to a flatter, more predictable state. These strategies are effective, but each spends additional actions and sensing before the grasp itself, and their reliability degrades exactly in the flat-cloth-on-surface configuration this thesis targets, where there is no wrinkle to find and no free edge to slide against.

2.2.2 Specialised End-Effectors

The second family redesigns the gripper so that a protruding feature is unnecessary. Monkman's survey of grippers for fibrous materials [14] catalogues the classical industrial approaches, developed largely for garment manufacturing: intrusive (needle/pin) grippers that pierce the top ply, and adhesive or electrostatic surface grippers. Electroadhesion has since matured substantially; Guo et al. [15] review its use across robotics and highlight its suitability for delicate, flexible materials at low energy cost. Non-contact airflow methods lift fabric without touching it at all: Ozcelik and Erzincanli's Bernoulli-principle end-effector [16] suspends a garment ply on a radial airflow film. More recently, variable-friction fingertips blur the line between gripper and manipulation policy: Jiang et al. [17] switch fingertip surfaces between high- and low-friction states to slide, pinch, and unfold fabric in-hand. These end-effectors trade generality for capability: needles risk damaging fine textiles, electroadhesion and airflow devices require dedicated power/pneumatic infrastructure and are largely single-purpose, and all of them abandon the ubiquitous two-jaw form factor that general-purpose robot platforms (including the Franka Hand used here) provide.

2.2.3 The Gap: the Gripper as an Experimental Variable

Across both families, the gripper hardware is treated as *fixed* and the innovation budget is spent on perception, policy, or a task-specific mechanism. Direct comparisons between compliant and rigid jaw-type grippers exist; Li et al. [18] systematically review soft–rigid hybrid grippers against industrial parallel-jaw grippers across payload, adaptability, and precision metrics. Such comparisons, however, are conducted on rigid or quasi-rigid workpieces, not on cloth, and report capability envelopes rather than parametric contact behaviour.

The closest precedent to this thesis is the household-clothing benchmark of Clark et al. [19], who evaluated several commercial end-effectors (including finray-style compliant fingers mounted on a Franka Emika Hand, alongside the standard Franka Hand, a Robotiq 2F-140, and the OpenHand Model T42) on four cloth-specific benchmarks: edge-drag accuracy, edge-grasp resilience, crumpled-object encapsulation, and fold generation. Their results establish empirically that end-effector selection materially changes cloth-manipulation outcomes, and their inclusion of finray fingers on the same robot platform used here makes the comparison direct. The evaluation, however, is scored at the *task* level: each benchmark reports success or accuracy of a completed manoeuvre, with the approach pose fixed per task. What it does not measure is how the grasp itself degrades as the approach conditions vary: the continuous dependence of contact force and grasp outcome on press depth and gripper tilt that any deployed system experiences as positioning error.

None of the work above, therefore, addresses the specific question this thesis investigates: cloth-manipulation literature treats grasp *success* as an outcome to be predicted or achieved at a fixed operating point, not as a function of a continuous physical parameter space (press depth, gripper tilt) whose boundary can be mapped and compared across gripper geometries. The benchmark in this thesis is deliberately narrower in manipulation scope than folding or dressing pipelines, but correspondingly more quantitative about the mechanics of the grasp itself: it keeps the two-jaw form factor fixed, varies only the finger design (t, α) (defined in Figure 3.1), and maps the resulting contact-force landscape under the positional and angular uncertainty (d, θ_x, θ_y) that any deployed system must tolerate.

2.3 Compliant Gripper Designs

The finray-effect finger design used in this thesis originates in a patent by Kniese [20], who observed that a structural element built from two flexible longitudinal members joined by hinged cross-struts deflects *toward* an applied lateral load rather than away from it, the counter-intuitive "bends toward the pressure" behaviour later commercialised by Festo as the Fin Ray Effect®. Festo's own current technical documentation for its DHAS gripper-finger product line [21] describes this structure explicitly as two flexible bands joined by cross-ribs at flex hinges, and reports measured retention-force, lateral-force, and indentation-depth curves as functions of gripping force and workpiece diameter. These are the same three quantities (force, indentation depth, contact geometry) that this thesis's depth-sweep experiments characterise directly on 3D-printed finray fingers rather than the commercial product.

Academic characterisation of the finray design has followed two directions: analytic modelling and empirical/manufacturing studies. Shan and Birglen [22] developed an analytic mechanical model of finray fingers to predict grasp force and finger deformation from geometric parameters, addressing the absence of a convenient design tool for this class of compliant finger. Their model, like the empirical studies that follow it, addresses *lateral* grasping contact; no published model, to the author's knowledge, predicts the axial tip-compression response (the rise–peak–valley–recovery profile) characterised empirically in this thesis. On the empirical side, Crooks et al. [23] modified the finray design by angling its internal crossbeams to introduce a directional bending bias, fabricated by multi-material 3D printing, and reported an approximately 40% increase in holding capacity over a traditional finray finger from this geometric change alone. Elgeneidy et al. [24] characterised fully 3D-printed finray fingers made from flexible thermoplastic polyurethane (TPU) filament, varying crossbeam rib angle and spacing, and found that at large tip displacements the ribs contact one another (*layer jamming*), substantially increasing force generation and producing a tunable-stiffness effect. Most directly relevant to this thesis's use of TPU-printed fingers, Lang et al. [25] 3D-printed finray fingers in both TPU 60A and TPU 95A and tensile-tested them: the softer 60A durometer gave the better force/compliance balance and left delicate fruit undamaged, while 95A reached substantially higher forces but visibly bruised the fruit. That result is direct prior evidence that finger material hardness, independent of finger geometry, is a first-order design lever for a compliant gripper, a finding consistent with this thesis's own manufacturing-variability results (Section 5.1).

The operational-envelope argument for compliant hands (that compliance buys tolerance to positioning error) was established quantitatively by Dollar and Howe's SDM Hand [26]: an underactuated hand with compliant polymer fingers that grasped a 48 mm cylinder at positions offset by more than 100% of the object size in each direction, while keeping acquisition contact forces low, where a rigid gripper would require millimetre-level placement. Their successful-grasp-range maps over an xy offset grid are the direct methodological ancestor of the depth–tilt working-window maps produced in this thesis, transposed from rigid household objects to cloth and from lateral offset to press depth and tilt. Closest of all in spirit, Teeple et al. [27] decomposed the benefit of a pneumatic soft gripper's compliance into per-axis contributions when handling thin, flexible materials: vertical compliance was shown to buy robustness to large vertical positioning uncertainty, measured head-to-head against a rigid gripper, while lateral and rotational compliance lengthen the response window and cap the tensile force during snags. Their vertical-uncertainty comparison is the single-gripper-pair analogue of this thesis's central question; the benchmark here extends it across a family of finger designs, adds the tilt dimension, and grounds it on cloth with a fully passive compliant finger.

Compliant gripping is not achieved solely through finray-style flexure. Granular jamming is the other major compliant-gripping paradigm in the literature: Brown et al. [28] demonstrated that a latex membrane filled with granular material conforms to an object's shape under positive pressure and rigidifies under vacuum, enabling passive universal grasping without dedicated fingers at all, and Amend et al. [29] extended this with a combined positive/negative pressure actuation cycle for faster, more reliable grasp-and-release cycles. A third paradigm, pneumatic elastomer actuation, underlies commercial soft grippers such as those from Soft Robotics Inc.; the technology traces to pneumatic-network (PneuNet) elastomer actuators developed by Ilievski et al. [30], and the commercial mGrip product line [31] demonstrates that compliant gripping has moved beyond academic prototypes into food-handling industrial deployment. This thesis's finray variants and rigid parallel-jaw comparator sit within this broader design space as one specific, geometrically simple point in it, chosen because (unlike jamming or pneumatic actuation) a finray finger requires no additional actuation hardware beyond the gripper's own jaw motion.

2.4 Force-Controlled Grasping

The theoretical basis for treating contact force, rather than position alone, as a control variable during manipulation is Hogan's impedance control framework [32, 33, 34]: neither pure position control nor pure force control is adequate for a task involving contact with an environment of unknown or varying mechanical properties, so the manipulator should instead regulate the dynamic relationship — the impedance — between its motion and the interaction force at its end-point. This thesis's trial procedure, which presses the gripper to a commanded depth and then reports the resulting *stationary* contact force F_z rather than commanding force directly, is an impedance-control-style measurement: depth is the commanded quantity, force is the observed response, and the relationship between them (Chapter 5) is exactly the impedance curve Hogan's framework predicts should characterise contact with an unknown environment (here, the cloth-covered rigid table).

On the Franka FR3 platform specifically, libfranka exposes two real-time command modes: external torque-level control (including a Cartesian impedance controller distributed as a reference implementation) and Cartesian pose motion generation, in which the client streams pose targets at the 1 kHz servo rate and tracking is delegated to the arm's internal impedance controller [35]. The tilt-aware pose targeting used in this thesis's trial procedure (Section 3.4) uses the latter mode: the in-house `fr3_pose_controller` commands Cartesian poses through libfranka's pose interface, so the contact interaction is mediated by the robot's internal impedance law rather than a user-tuned stiffness.

Contact force in this thesis is measured through the Franka FCI's own external-wrench estimate; a second, independent channel using an array of Tekscan FlexiForce resistive force sensors was provisioned, but a repeatable calibration could not be established in time and the cross-check is carried as future work (Chapter 7). FlexiForce is a thin-film piezoresistive sensor marketed explicitly for grip-force measurement in robotic and surgical applications [36], and has prior academic precedent for exactly this use: Sharan and Onwubolu [37] instrumented a two-finger pick-and-place gripper with a FlexiForce sensor to measure grip force directly against known payload weight. An alternative tactile-sensing approach not used in this thesis but relevant for future work is vision-based tactile sensing: Yuan et al.'s GelSight [38] sensor measures the deformation of a soft elastomer contact surface via an internal camera, recovering high-resolution contact geometry from which force and slip can be inferred, in contrast to FlexiForce's direct (but lower-resolution) force read-out.

Finally, the trial procedure's touch-off step (descending the gripper under position control until a chosen contact criterion is met, before switching to the pressed-depth grasp) follows a long lineage of contact-detection strategies in manipulation. Whitney's classical analysis of quasi-static assembly [39] derives the geometric and force-equilibrium conditions under which measured contact force can be used to detect and guide compliant insertion, the analytical basis for the Remote Center Compliance device and, more generally, for using force as a contact-detection signal rather than only as an outcome measure. De Luca et al. [40] showed that contact can be detected from proprioceptive joint measurements alone, without a dedicated force/torque sensor, via a generalised-momentum residual, a capability this thesis's FCI-based touch-off implicitly relies on when it uses the controller's own external-wrench estimate rather than an external load cell.

2.5 The Franka FR3 Platform

The Franka Research 3 (FR3) is a 7-degree-of-freedom, torque-controlled collaborative arm with seven integrated joint torque sensors, a 3 kg payload, and position repeatability under ±0.1 mm, accessed in this thesis through the Franka Control Interface (FCI) [41, 42], a real-time control layer providing torque, position, and velocity access in both joint and Cartesian space at up to 1 kHz. This thesis controls the arm through `franka_ros2` [43], the official ROS 2 integration built on `libfranka` and `ros2_control`.

The FR3's predecessor, the Franka Emika Panda, is documented by its manufacturer as a widely adopted reference platform for robotics research and education [44], and a more recent survey by the same author quantifies its continued adoption as a de facto standard platform across manipulation, control, and human-robot-interaction research [45]. That widespread adoption has produced independent characterisation studies directly relevant to this thesis's instrumentation choices. Most pertinently, Petrea et al. [46] empirically evaluated the accuracy of the Panda's built-in filtered external-force estimate against ground-truth load-cell measurements during controlled contact. This is the same FCI external-wrench signal this thesis uses as its Contact Force channel, so their accuracy findings directly bound how much that channel can be trusted, and motivate the independent force-sensor cross-check identified as future work. Gaz et al. [47] separately derived a full dynamic (inertial and friction) parameter model of the Panda's arm via penalty-based optimisation, a model since reused widely enough to be treated as a de facto standard dynamic model of the platform. Beyond force sensing specifically, the Franka Panda/FR3's broad adoption as a manipulation-research testbed is illustrated by work such as Chi et al.'s [48] diffusion-policy visuomotor control framework, evaluated on contact-rich real-world manipulation tasks using a Franka Panda as one of its physical platforms. This adoption is directly relevant to the control mode chosen here: learned visuomotor policies of this kind emit streams of end-effector pose targets, the same command interface through which this thesis's trial procedure drives the arm (Section 3.4). The benchmark is therefore conducted through the interface a learned or simulation-trained policy would use in deployment, with sim-to-real transfer as the motivating trajectory: a compliant finger that tolerates centimetre-scale press-depth error and degree-scale tilt error widens the placement-error budget available to a policy trained in simulation, and the depth–tilt maps produced in this thesis quantify that budget as a function of gripper design.

2.6 Position and Summary

This thesis sits at the intersection of three threads: cloth-handling system design, finray-effect gripper characterisation, and force-controlled manipulation on collaborative robots. Prior work has independently established each thread: cloth manipulation research has largely optimised for task-level success (folding, singulating, dressing) under vision-driven or learned control; compliant-gripper research has characterised finray and jamming designs primarily through force/holding-capacity metrics on rigid or quasi-rigid test objects; and impedance-control theory and its Franka FCI implementation provide the tools to regulate and measure contact force but have not, in the reviewed literature, been used to map how that force varies over a continuous depth–tilt parameter space for a fragile, non-rigid workpiece. The contribution of this thesis is a unified experimental treatment of all three on a single platform with consistent instrumentation, enabling direct quantitative comparison between rigid and compliant gripper response under deliberately perturbed conditions.

3

Experimental Setup and Methodology

3.1 Hardware Configuration

3.1.1 Robot Manipulator

A Franka Research 3 (FR3) 7-DOF cobot served as the manipulator throughout this work. The arm was operated through the Franka Control Interface (FCI) using `libfranka 0.18.1` and the ROS 2 Humble control stack. Cartesian pose targets were streamed through libfranka's motion-generator interface by a custom `fr3_pose_controller` developed and maintained by the McGill Applied Dynamics Lab, with tracking delegated to the arm's internal impedance controller (Section 2.5). The controller accepts a sequence of (pose, twist) waypoints with associated time stamps and interpolates between them using a quintic polynomial; for the in-place re-orientation and the final precision descent phases, a cosine half-cycle S-curve interpolator (the "sine descent") is used to ensure zero start- and end-state velocities.

3.1.2 Gripper Variants Tested

Table 3.1: Six gripper variants benchmarked, indexed by blade thickness t and baseline seat-mount tilt α (Figure 3.1). All finray variants share the same 3D-printed inclined-seat mount; only α changes between the $+2.5°$ baseline and the $+5.0°$ variant. The parallel-jaw variants do not use an inclined seat and have no defined t or α.

#	Variant	Type	Finger	t	α
1	`parallel_jaw_stock`	Rigid	Metal jaws (Franka Hand)	—	—
2	`parallel_jaw_TPU`	Rigid + pad	2.0 mm TPU 90A pads	—	—
3	`finray_18`	Compliant	TPU 90A print	18 mm	+2.5°
4	`finray_18_model2`	Compliant	Second print of #3	18 mm	+2.5°
5	`finray_26`	Compliant	TPU 90A print	26 mm	+2.5°
6	`finray_26_5deg`	Compliant	TPU 90A print, 5.0° seat	26 mm	+5.0°

The finray fingers and the parallel-jaw compliant pad were 3D-printed from Bambu Lab TPU 90A (Shore-A hardness 90) on a Bambu Lab X1 Carbon printer using the vendor-default TPU 90A profile in Bambu Studio (0.4 mm nozzle, default layer height, default infill, PEI-textured build plate).

Baseline inclined-seat mount. We define α as the baseline seat-mount tilt angle, annotated in the front view of Figure 3.1: the fixed rotation about the gripper's local y axis (the gripping direction) introduced by the 3D-printed seat adapter that carries every finray finger. The figure shows the mounting stack (flange → seat → finger) and the α and t design-parameter convention. All finray variants were mounted on the Franka Hand via this inclined seat; the seat was present in *every* finray variant tested in this thesis unless otherwise noted, and is the physical realisation of α. Variants whose name contains `_5deg` used an $\alpha = +5.0°$ seat in place of the default $\alpha = +2.5°$ seat; all other finray variants (`finray_18`, `finray_18_model2`, `finray_26`) used the $\alpha = +2.5°$ baseline; the parallel-jaw variants have no seat and no defined α. The $\alpha = +2.5°$ baseline was chosen so that the two opposing finray fingers converge inward toward each other rather than splaying apart. This closes the residual slit that would otherwise remain between the finger tips when the gripper is fully closed, and ensures firm contact along the fingers' full length during a successful grasp.

Choice of design-parameter values. The two blade thicknesses and the second seat angle were not arbitrary. The 26 mm blade matches the finger dimension of the handheld Universal Manipulation Interface (UMI) gripper [49], whose fin-ray-type soft fingers have become a de facto reference design for learned manipulation, so results at $t = 26$ mm transfer directly to that widely used hardware. The 18 mm blade matches the thickness of the stock Franka parallel jaw, giving a compliant finger and a rigid finger of equal frontal dimension for the cross-class comparison. The $\alpha = +5.0°$ seat was chosen to sit outside the $\pm 2.5°$ runtime tilt range: with a seat at the baseline $+2.5°$, a runtime tilt of $-2.5°$ would exactly cancel the seat inclination and confound the two effects, whereas the 5.0° seat keeps the fixed mount contribution distinct from every runtime tilt tested.

Consequence for reported "0°" cells. Cells reported in Chapter 5 as 0° tilt denote that the robot arm is perpendicular to the surface ($\theta_x = \theta_y = 0$; the runtime tilt angles are illustrated in Figure 3.3). Runtime tilt values ($\pm 2.5°$ about x or y) tilt the arm away from this perpendicular pose.

3.1.3 Cloth Specimen

All trials used a single cloth specimen: a 30×22.5 cm piece of single-layer plain-woven cotton (Figure 3.2). The same physical specimen was reused for the entire campaign and reset to a flat, wrinkle-free configuration before every trial (Section 3.6, step 0), so that each trial began from the flat-on-surface configuration this thesis targets.

3.1.4 Test Surface

The cloth rested on a rigid, flat acrylic plate that served as the test surface for the entire campaign, fixed to the workbench so that its position and orientation were common to all variants and sessions. The surface height entering every trial was not assumed but measured: the force touch-off of Section 3.2.2 re-detected the surface at each calibration, making the plate's absolute height a per-session measured quantity rather than a fixed constant.

[TODO: insert photograph of the test bench in situ and short caption describing the table material, flatness, and cloth-edge orientation relative to the gripper y axis.]

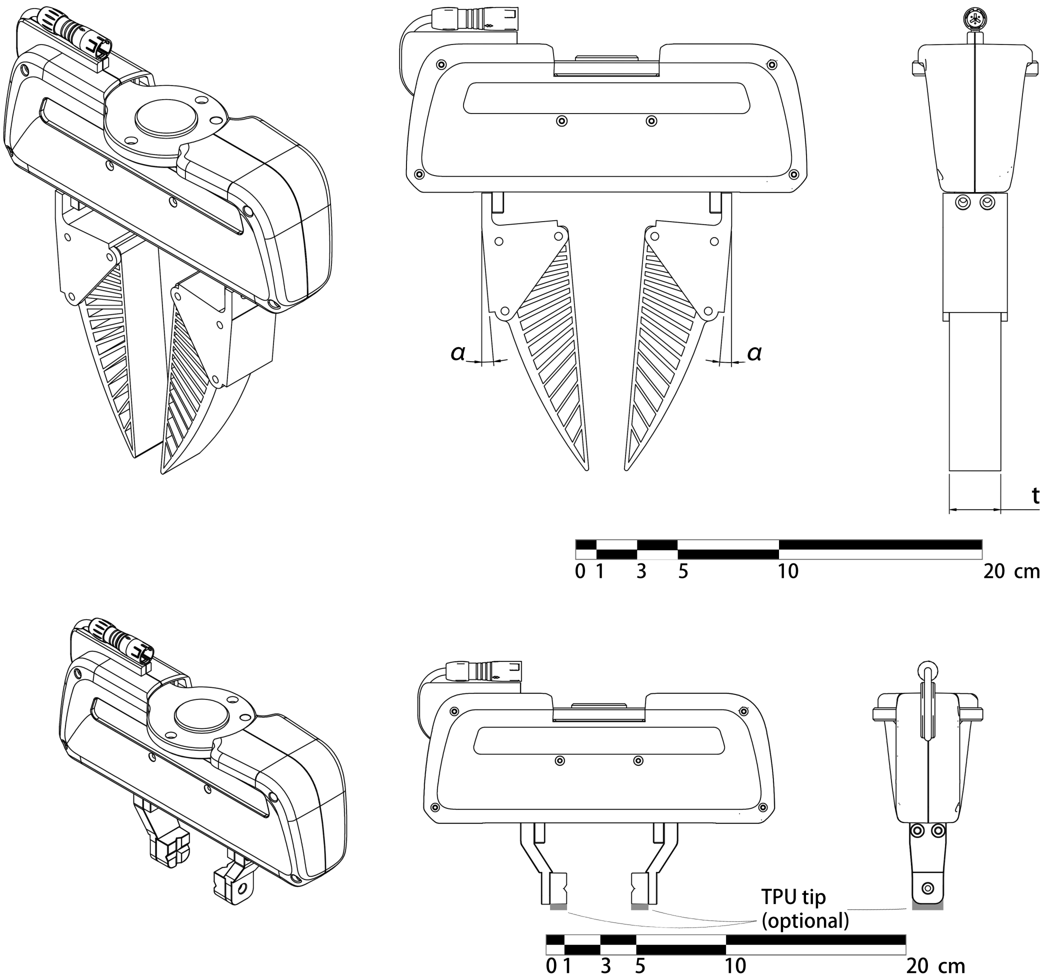


Figure 3.1: Both grippers in isometric, front, and side orthographic views, mounted on the Franka Hand. Top: a finray variant; the front view (centre) annotates the baseline seat-mount tilt α at each finger root, and the side view (right) annotates the finger blade thickness t. Bottom: the rigid parallel-jaw gripper with the optional TPU tip indicated. Scale bars in cm.

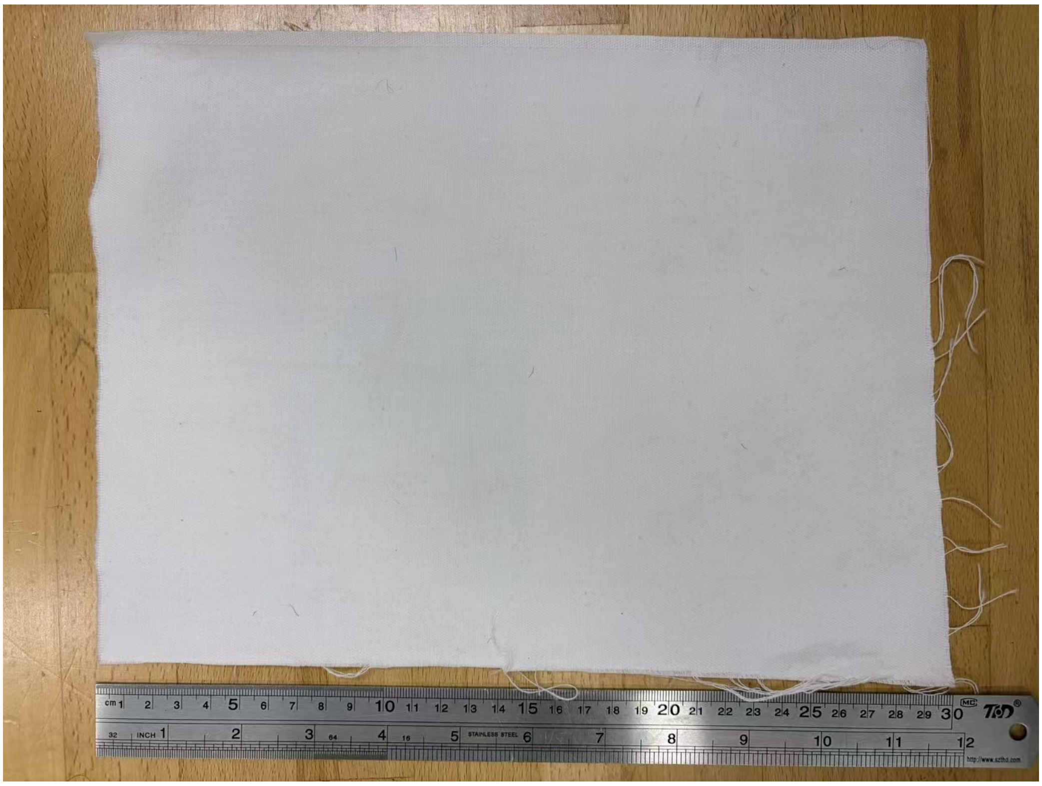


Figure 3.2: The single-layer plain-woven cotton specimen used throughout this thesis, $30 \times 22.5\,\mathrm{cm}$, shown against a ruler for scale.

3.2 Reference Frame and Tool-Centre Calibration

3.2.1 Frame Convention

Three frames are relevant: the world (robot base) frame $\mathcal{W}$, the gripper flange frame $\mathcal{F}$, and the gripper tip frame $\mathcal{T}$. The Franka URDF defines $\mathcal{F}$ at the mechanical hand mounting flange; the Franka API `robot.end_effector_pose` reports the pose of $\mathcal{F}$ in $\mathcal{W}$. The tip frame $\mathcal{T}$ is located along the local $+z$ axis of $\mathcal{F}$ by a per-variant offset z_{TCP}, which depends on the physical gripper (finger length, mounting fixtures, pads).

Throughout, $\mathbf{p} \in \mathbb{R}^3$ (boldface) denotes a position vector expressed in $\mathcal{W}$ unless otherwise noted, and $R_x(\phi)$, $R_y(\phi)$ denote the elementary rotation matrices for a right-handed rotation by angle ϕ about the fixed world x or y axis, so that a general gripper orientation is a composition $R \in SO(3)$ of these primitives. For a gripper at orientation R, the world-frame relationship between the tip and flange positions is

$$\mathbf{p}_{\mathcal{T}} = \mathbf{p}_{\mathcal{F}} + R \begin{bmatrix} 0 \\ 0 \\ z_{\mathrm{TCP}} \end{bmatrix}. \tag{3.1}$$

For the rigid-down-pointing nominal orientation $R_0 = R_x(180°)$, the tip lies directly below the flange and $\mathbf{p}_{\mathcal{T}} = \mathbf{p}_{\mathcal{F}} - z_{\mathrm{TCP}}\hat{\mathbf{z}}$.

The gripper-frame convention adopted here is: the gripping axis (along which the fingers close) is $\mathcal{F}$'s local y axis; the gripper is mirror-symmetric about the xz plane; consequently, rotation of the gripper about the local y axis is rotation about the geometrically asymmetric axis, while rotation about x is rotation about the symmetric axis. Because the nominal orientation $R_0 = R_x(180°)$ is itself a rotation about the world x axis, the world x and y axis *lines* coincide with the flange's local symmetric and gripping axis lines at R_0 (up to sign). The runtime tilt angles θ_x, θ_y introduced in Section 3.4 (Figure 3.3) are defined as rotations about these fixed world axes, and because the axis lines coincide with the gripper-local axes at R_0, the same symmetric-axis / gripping-axis language applies without ambiguity to θ_x and θ_y respectively.

3.2.2 TCP Offset Calibration

The z_{TCP} value differs by variant because different fingers attach to the same flange. To calibrate, each variant was driven to a common probe position (x_0, y_0) on a known rigid table and a force-feedback touch-off was performed: the gripper descended slowly in z until the external z force exceeded 3 N in magnitude (debounced for three consecutive 50 Hz samples). The flange z at the latch moment was recorded as $z_{\text{TCP}}^{\text{measured}}$.

Taking the rigid `parallel_jaw_stock` variant as the URDF reference ($z_{\text{TCP}} \equiv 0$ by definition), the per-variant TCP offset is

$$z_{\text{TCP}}^{(v)} = z_{\text{TCP}}^{(v),\text{measured}} - z_{\text{TCP}}^{\text{stock,measured}}. \tag{3.2}$$

Table 3.2 summarises the calibrated values used throughout this work.

Table 3.2: Calibrated per-variant TCP offsets from differential touch-off against the stock parallel jaw. Offsets are computed from the unrounded calibration records; displayed values are rounded independently. `finray_18_model2` is absent because it was re-calibrated on mounting (see text).

Variant	$z_{\text{TCP}}^{\text{measured}}$ (mm)	$z_{\text{TCP}}^{(v)}$ (mm)
`parallel_jaw_stock`	37.871	0.000
`parallel_jaw_TPU`	39.916	2.044
`finray_18`	131.842	93.971
`finray_26`	132.782	94.911
`finray_26_5deg`	130.094	92.223

The reproducibility of the touch-off was measured as the across-tap standard deviation $\sigma_{z,\text{tcp}}$ in mm (not to be confused with the per-cell force standard deviation σ_{F_z} of Section 3.7). For `finray_26_5deg`, $\sigma_{z,\text{tcp}} = 183\,\mu\text{m}$ owing to a wobble component introduced by the inclined seat; the other four variants had $\sigma_{z,\text{tcp}} \leq 102\,\mu\text{m}$. The second print (`finray_18_model2`) shares the finger geometry and seat of `finray_18`; its surface reference and TCP offset were re-established by the same touch-off procedure when it was mounted, rather than reusing the first print's calibration.

3.3 Force Measurement

External forces $\mathbf{F}_{\text{ext}}$ at the flange were estimated by the Franka internal model from joint torques, available via the `libfranka` RobotState API at 1 kHz and subsampled to 30 Hz for session logging. A free-air bias capture was performed at the start of every session: the gripper was held at a known pose 30–80 mm above the cloth surface and 30 samples were averaged over 1 s to provide F_z^{bias}. This bias was recorded in each session JSON but *not* subtracted from the raw force time series at acquisition time; downstream analyses subtract it where appropriate.

3.4 Tilt-Aware Pose Targeting

When the gripper is tilted, the flange position required to place the tip at a fixed probe location $(x_0, y_0, z_{\text{surf}})$ shifts laterally. From equation (3.1), the required flange position is

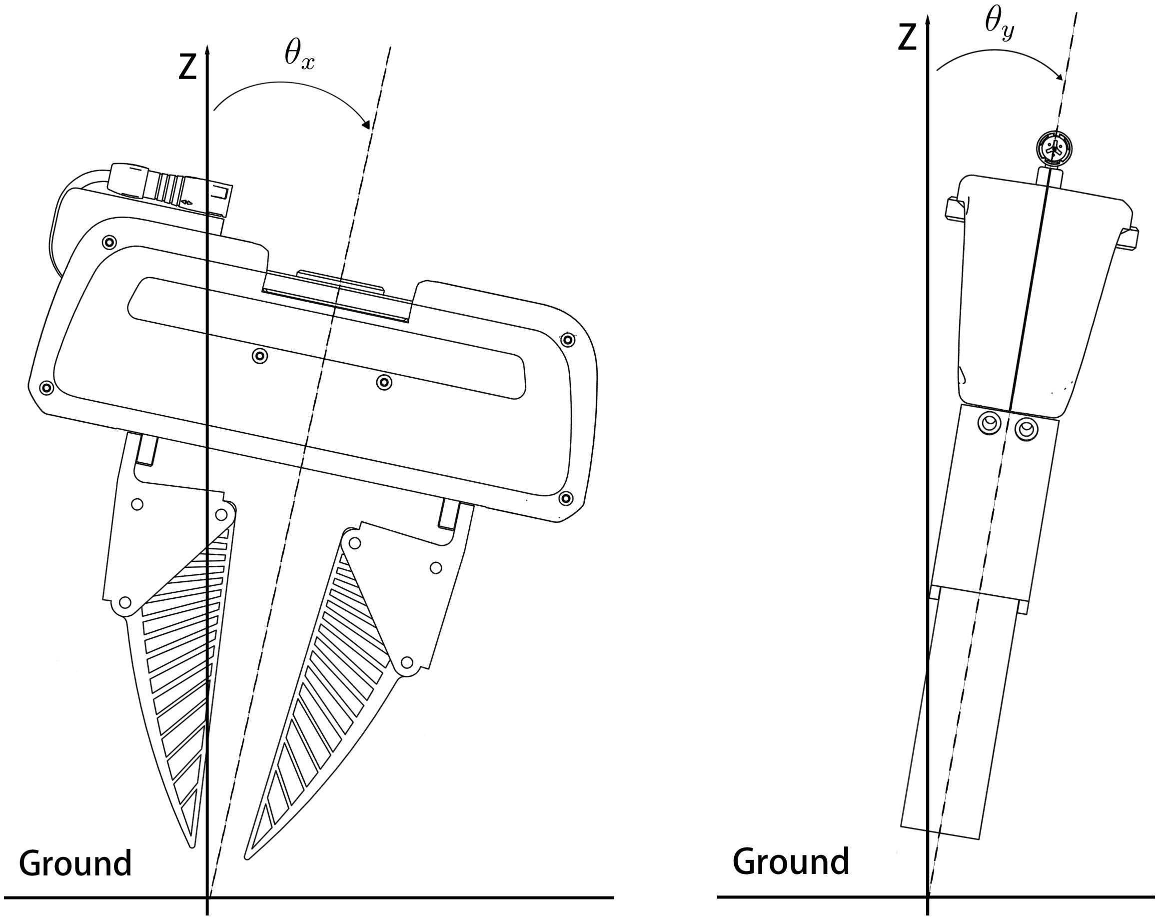
$$\mathbf{p}_{\mathcal{F}}^{*} = \mathbf{p}_{\mathcal{T}}^{\text{target}} - R \begin{bmatrix} 0 \\ 0 \\ z_{\text{TCP}} \end{bmatrix}. \tag{3.3}$$

This relation is implemented in the trial-control software: the target orientation $R = R_{\text{tilt}} \circ R_0$ is constructed at session start by composing the requested tilt rotation $R_{\text{tilt}} = R_y(\theta_y) R_x(\theta_x)$ (extrinsic, world frame) onto the base orientation $R_0 = R_x(180°)$, and the flange position target is updated accordingly. The orientation is then held constant through all subsequent descent and grasp phases, so the tip XY remains at (x_0, y_0) throughout the trial. Figure 3.3 illustrates the two runtime tilt angles: θ_x rotates the gripper about the symmetric axis (front view, both fingers move together), while θ_y rotates it about the gripping axis (side view, the mirror-symmetric fingers overlap in projection). Both are shown here as a general rotation from the world z axis ("Ground"); the runtime sweep used in this study is $\theta_x, \theta_y \in \{-2.5°, 0°, +2.5°\}$. These arm rotation angles are a separate quantity from the seat angle α of Section 3.1.2: α is a fixed property of the gripper hardware (how the fingers meet the flange), while θ_x, θ_y describe how the arm is oriented relative to the surface. The two are reported independently throughout this thesis and are never summed.

For a $+2.5°$ tilt about x on the `finray_26` variant ($z_{\mathrm{TCP}} = 94.911\,\mathrm{mm}$), the predicted flange offset is

$$\Delta y_{\mathcal{F}} = -z_{\mathrm{TCP}} \sin(2.5°) \approx -4.14\,\mathrm{mm}. \tag{3.4}$$

Chapter 4 reports the empirical lateral offsets measured under load and the residuals from this geometric prediction, which we attribute to finger compliance under press.



(a) θ_x: rotation about the symmetric axis (front view)

(b) θ_y: rotation about the gripping axis (side view)

Figure 3.3: Runtime tilt angles θ_x (left) and θ_y (right), each shown as a rotation of the whole gripper body away from the world z axis (labelled) relative to the Ground line; 0° is the arm perpendicular to the surface.

3.5 Data Pipeline

3.5.1 Per-Trial Recording

Each trial directory `trial_NNN/` contains:

- `config.yaml`: snapshot of the variant config at trial time;
- `run_config.json`: depth, tilt, ground-truth source, speeds, gripper mode;
- `pose_wrench.csv`: high-rate (30 Hz) actual + commanded pose, RPY, wrench, gripper width per row, tagged with the current trial stage.

3.5.2 Per-Session Recording

Each session directory `depth_*/` contains in addition:

- `depth_summary.json`: header metadata + array of per-trial verdicts (`success`, `fail_reason`); operator notes;
- `ground_truth_used.json`: snapshot of the GT JSON loaded;
- `session_log_*.npz`: numpy archive of the 30 Hz background sample stream, with stage labels and tap indices.

3.5.3 Verdict Recording and Outcome Classification

Trial outcomes were classified into three categories rather than a binary success/failure. After each trial the operator was prompted to distinguish *success* (the gripper held the cloth throughout the lift stage) from *no-grasp* (the trial executed to completion and the stationary force sample was valid, but the cloth slipped from the gripper during or after the lift). Trials in which the FCI control daemon aborted the motion mid-trial were recorded as *failed trials* and excluded from all aggregate analyses; the two most common triggers were:

- the Franka safety reflex firing on excessive predicted joint force (`auto_failed_excess_force`), which typically occurred when the parallel-jaw variants were commanded to depths beyond their narrow force-limited working range;

- the Franka safety reflex firing on acceleration discontinuity between two consecutive Cartesian trajectory segments, which was observed intermittently during the chunked touch-off descent and mitigated by raising the trajectory-emission floor of the touch-off descent trajectories.

Both trigger classes left the trial's `success` field as `null` in the per-trial JSON; no-grasp trials, by contrast, carried `success = false` with a valid force record and were retained for engagement-force analysis.

3.6 Trial Procedure

A single trial proceeded as follows:

0. Cloth reset: the operator flattened the cloth on the test surface, smoothing out any wrinkles or folds left by the previous trial, so that every trial began from the flat-on-surface configuration the benchmark targets;

1. The gripper opened (Franka Hand or finray release width);

2. Bulk descent: a smooth multi-waypoint trajectory carried the gripper from the lifted pose down to a fine clearance (+5 mm above surface), filtered to exclude any upward leg (preventing direction reversals at zero-twist interior waypoints);

3. Settle: a 2 s pause at the fine-clearance pose;

4. Final approach: a cosine-S-curve descent carried the gripper from the fine clearance to the press target $z_{\text{surf}} - d$, where d is the trial press depth (Figure 3.4);

5. Grasp: a force-controlled close (Franka grasp action at 80 N or 20 N nominal, with ±100 mm ϵ to disable the position-acceptance check that would otherwise cause auto-release on a sub-spec final width);

6. Settle: a 1 s pause;

7. Stationary sample: 30 samples of $\mathbf{F}_{\text{ext}}$ were captured over 1 s during the `grasp_stationary_sample` stage;

8. The gripper lifted to $\sim +80$ mm above surface for visual inspection;

9. The operator was prompted for a verdict;
10. The gripper opened and the rig was reset for the next trial.

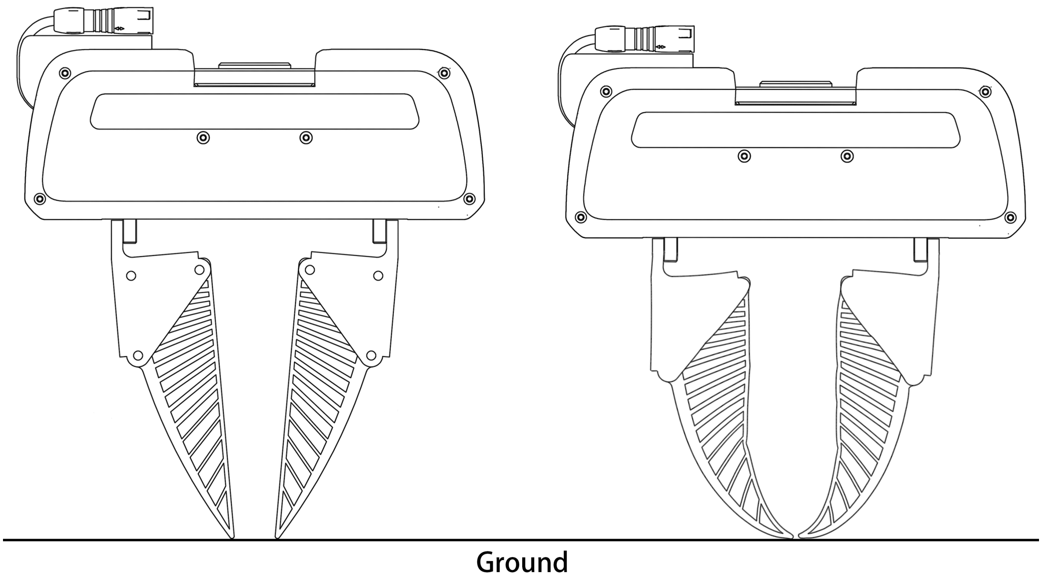


Figure 3.4: Physical effect of the press depth d on a finray variant. At first contact ($d = 0$, left) the blades are straight and only the tips touch the surface; pressed to $d > 0$ (right), the compliant blades buckle inward and wrap the surface, increasing the contact area. This wrap-around engagement underlies the depth-gated tilt robustness discussed in Chapter 6.

3.7 Statistical Treatment

3.7.1 Calibrated Depth Reference

All aggregate analyses are reported on a *calibrated* depth axis

$$d_{\text{cal}} = d - d_{\text{first success}}, \tag{3.5}$$

where $d_{\text{first success}}$ is, per trial series (each variant–tilt-condition pair, within a calibration epoch), the shallowest cell in which at least two-thirds of the valid completed trials succeed (the majority-success criterion); each series is zeroed at its own first majority-success cell. The reference is thus a directly measured physical event, the onset of reliable grasping, rather than the force-touch-off zero, which inherits cross-session bias drift and per-print geometry offsets (Section 3.2.2). Aggregate statistics and figures use only $d_{\text{cal}} \in [0, 10]$ mm; cells outside this window remain in the record but are excluded from aggregates. The 10 mm bound was chosen because, at that width, every finray series' record extends at least to the window end, so every finray range in Chapter 5 is *right-censored* uniformly: the true working range is known only to be at least 10 mm, since the underlying record continues beyond the window but is excluded from the aggregate by convention, not because the gripper stopped working there. The bound also keeps every finray's peak–valley shape (Section 5.5) fully contained within the window. A wider window (e.g. 12 mm) was evaluated and rejected: one finray variant's shape curve extended only to 10.5 mm of usable support at that width, breaking the uniform-censoring property. The rigid parallel-jaw variants' working ranges are unaffected by the window choice, as both lie entirely inside $[0, 10]$ mm and their ranges are therefore reported as closed measurements, not censored. This convention was adopted as Bill 0007 of the project's governance record (enacted 2026-08-09; revised through 2026-08-10 to per-series zeroing, a majority-success cell criterion, and the final 10 mm window), on the evidence that first-success alignment collapses the two prints of the 18 mm finger onto a common curve.

3.7.2 Per-Cell Statistics

For each (variant, depth, tilt) cell, the contact force is reported as

$$\bar{F}_z = \frac{1}{N}\sum_{i=1}^{N} F_z^{(i)}, \qquad \sigma_{F_z} = \sqrt{\frac{1}{N}\sum(F_z^{(i)} - \bar{F}_z)^2}, \tag{3.6}$$

where N is the total number of stationary samples pooled across all clean trials in the cell. Trials with null verdict (FCI crash) are excluded. Trial-to-trial variance is the dominant source of dispersion; the within-trial sample variance (~ 0.1 N) is small compared to it.

4

Experimental Design and Procedure

The empirical campaign was organised into two sequential phases on a single experimental apparatus. Phase 1 established the contact-force versus press-depth relationship for each gripper variant at nominal zero-tilt orientation, and Phase 2 perturbed the same measurement with $\pm 2.5°$ tilts about the two orthogonal Cartesian axes. Together the two phases define a three-dimensional (depth, tilt-axis, tilt-sign) cell grid whose per-cell measurements constitute the data corpus reported in Chapter 5.

4.1 Study Design

The gripper roster comprised six variants organised into two families:

- **Finray family** (four variants): `finray_18`, `finray_18_model2` (second print of the same CAD), `finray_26` (larger blade), and `finray_26_5deg` (`finray_26` on a 5°-inclined seat).

- **Parallel-jaw family** (two variants): `parallel_jaw_stock` (bare rigid jaws) and `parallel_jaw_TPU` (rigid jaws with a 2 mm TPU 90A compliant pad).

The same fabric specimen (single-layer cotton) rested on the same acrylic table surface throughout the campaign. The Franka FR3 arm held the gripper in a fixed base orientation $R_0 = R_x(180°)$ for Phase 1, and applied signed rotations $R_\theta = R_x(180°)\,R_y(\theta_y)\,R_x(\theta_x)$ for Phase 2, with $(\theta_x, \theta_y) \in \{(\pm 2.5°, 0), (0, \pm 2.5°)\}$ (the runtime arm tilts of Chapter 3, Figure 3.3, distinct from the seat angle α of Figure 3.1).

The perturbation magnitudes were chosen to represent the placement uncertainty of a deployment that cannot afford precise calibration: a low-accuracy or coarsely mounted robot arm may plausibly carry an orientation error of a few degrees and a surface-height error of several millimetres. The $\pm 2.5°$ runtime tilt stands in for that orientation error, and the depth sweep (with its 10 mm calibrated analysis window, Chapter 3) spans the corresponding height-error budget. A gripper that tolerates the full tested range therefore tolerates a deployment in which neither the mounting angle nor the surface height is precisely known. Table 4.1 summarises the test conditions common to both phases.

Table 4.1: Test conditions of the experimental campaign. Symbols are defined in the Notation section; θ_x, θ_y are illustrated in Figure 3.3.

Variable	Definition	Values / precision
Press depth d	Fingertip driven below the detected surface	rigid: -1 to $+4$ mm at 0.25–0.5 mm steps finray: -2 to $+16$ mm at 0.5–1.0 mm steps
Tilt θ_x	Arm leaned about the symmetric axis	$-2.5°$, $0°$, $+2.5°$
Tilt θ_y	Arm leaned along the gripping direction	$-2.5°$, $0°$, $+2.5°$
Surface reference	Force touch-off at 3 N threshold	repeatable to ± 0.1 mm
Repeats	Trials per (variant, depth, tilt) cell	3

4.2 Depth Grid

The depth axis is defined relative to the per-session detected cloth surface via the `ground_truth_finder` touch-off procedure described in Chapter 3. The depth grid was chosen adaptively per gripper family:

- For the rigid parallel jaws, depths were spaced at 0.25–0.5 mm intervals over -1 to $+4$ mm, close to the narrow force-limited working range.
- For the finrays, depths were spaced at 0.5–1.0 mm intervals over -2 to $+16$ mm, covering the four regimes (boundary, knee, valley, recovery) identified in Chapter 5.

Per (variant, depth, tilt) cell, three trials were collected. Boundary depths were routinely retested when within-cell standard deviation exceeded 1 N, up to a maximum of two additional trials per (depth, tilt) combination.

4.3 Tilt-Aware Pose Geometry: Verification

Before reporting force results, the implementation of the tilt-aware tool-centre-point (TCP) tracking equation was verified empirically. Table 4.2 shows the measured flange XY shift during the stationary grasp sample versus the geometric prediction for `finray_26` at four representative (depth, tilt) cells.

Table 4.2: Measured versus predicted flange-XY offset during the stationary grasp sample at four cells. Predicted offset uses $z_{\text{TCP}} = 94.911$ mm.

Cell	Tilt cmd	Predicted $\Delta\mathbf{p}_{\mathcal{F}}$ (mm)	Measured $\Delta\mathbf{p}_{\mathcal{F}}$ (mm)	Residual	Mean F_z
-1.0 mm	$+2.5°$ y	$(+4.14, +0.00)$	$(+4.35, -0.03)$	0.21 mm	-8.00 N
-1.0 mm	$-2.5°$ y	$(-4.14, +0.00)$	$(-4.02, -0.04)$	0.13 mm	-3.39 N
$+10.0$ mm	$+2.5°$ y	$(+4.14, +0.00)$	$(+5.35, -0.28)$	1.24 mm	-36.26 N
$+10.0$ mm	$-2.5°$ y	$(-4.14, +0.00)$	$(-2.93, -0.29)$	1.24 mm	-35.95 N

At boundary depth, the residual was ≤ 0.21 mm, essentially the lateral positioning noise of the arm under light load. At safe depth, both positive and negative tilts exhibited a $+1.2$ mm common-mode offset in the same $+x$ direction, regardless of tilt sign. We interpret this common-mode offset as evidence of finger compliance under load: the ~ 36 N axial press deflects the soft finray blade laterally by approximately 1.2 mm in a structurally preferred direction. The differential offset (between $+y$ and $-y$ tilts) remains correctly predicted by the geometric model in both depth conditions.

This verification confirms that the control-frame implementation of the tilt-aware TCP tracker is correct, and any asymmetries observed in the contact force reflect physical gripper behaviour rather than control-system errors.

4.4 Rotation Tracking Verification

A natural concern is whether the Franka arm faithfully realises the commanded tilt orientation. The 30 Hz session log records actual and commanded RPY at every sample, enabling a direct measurement of orientation tracking error

$$\delta(t) = \|\mathrm{RPY}_{\mathrm{actual}}(t) - \mathrm{RPY}_{\mathrm{cmd}}(t)\|_2. \tag{4.1}$$

Across five representative sessions, the mean steady-state δ was 0.05° with a worst-case 0.13°. Transient excursions up to $\sim 5^\circ$ occur exclusively within the `orientation_lock` stage at session start, before any descent or grasp, and are filtered out of trial-level analyses.

A 2.5° commanded tilt is therefore $50\times$ the noise floor of the arm's rotation tracking; the observed force differences cannot be attributed to arm rotation error.

4.5 Trial Protocol and Outcome Classification

Each trial followed the fixed stage sequence detailed in Section 3.6: set the commanded pose above the target, descend to the pre-contact height, contact and press to the target depth, grasp with the gripper's default close command, lift by a fixed clearance, and record the operator's held/not-held verdict. A stationary-sample window of 1 s (30 samples) during the grasped-and-pressed stage provided the force measurement.

Trial outcomes were classified into three categories:

Success: the trial executed to completion, contact force was sampled cleanly during the stationary window, and the cloth remained gripped after the lift stage.

No-grasp: the trial executed to completion and contact force was sampled cleanly during the stationary window, but the cloth slipped from the fingers during or after the lift stage. The force measurement remains valid as an *engagement-force* characterisation.

Failed trial: the trial did not execute to completion. The controller either aborted with an excess-force safety event or the operator interrupted before the stationary window completed. The measurement is invalid and the trial is excluded from analyses.

The rationale for this three-way taxonomy, rather than a binary success/failure classification, is discussed in Section 6.2.

4.6 Experiment 3: Inclined-Surface Demonstration (Pilot)

Experiments 1 and 2 characterise the contact phase on a flat, levelled surface, with each variant calibrated against that surface. Experiment 3 tests the deployment consequence those measurements predict: when the surface geometry departs from what the calibration assumed, does the compliant finger keep grasping where the rigid jaw fails, with no change to calibration, policy, or sensing?

The planned protocol mirrors Experiments 1 and 2 in every respect except the surface and the outcome metric. The acrylic test surface is mounted on a wedge (`SURFACE_RISER`) whose CAD model specifies a top-surface incline of $\beta = 4.15°$ (measured from the model geometry: the top face normal is 4.1467° from vertical), with the slope aligned with the gripping direction; the as-built fixture will be verified against this nominal value before the session. Each variant is then calibrated *blind* by the standard force touch-off of Section 3.2.2, exactly as if the surface were flat and level: no tilt compensation is applied and the incline is treated as unknown to the calibration. Each trial presses to the calibrated touch-off depth plus 0.5 mm, the shallowest depth at which every variant grasped successfully on the flat surface, and the outcome is recorded as a binary grasp verdict under the outcome taxonomy of Section 4.5. Contact force is logged but is not a reported metric for this experiment.

The pilot covers all six gripper variants of Table 3.1, at three attempts each (18 trials total). From the flat-surface working ranges of Chapter 5, the registered prediction is that all four finray variants grasp successfully at every attempt, while both rigid variants fail or are marginal: a 4.15° surface-relative misalignment exceeds the largest runtime tilt at which the rigid jaws retained a usable window (2.5°), and that window had already narrowed to between 0.5 and 2.5 mm under that tilt, with one `parallel_jaw_stock` tilt condition reduced to a single successful depth and two producing no success at all (Section 5.3).

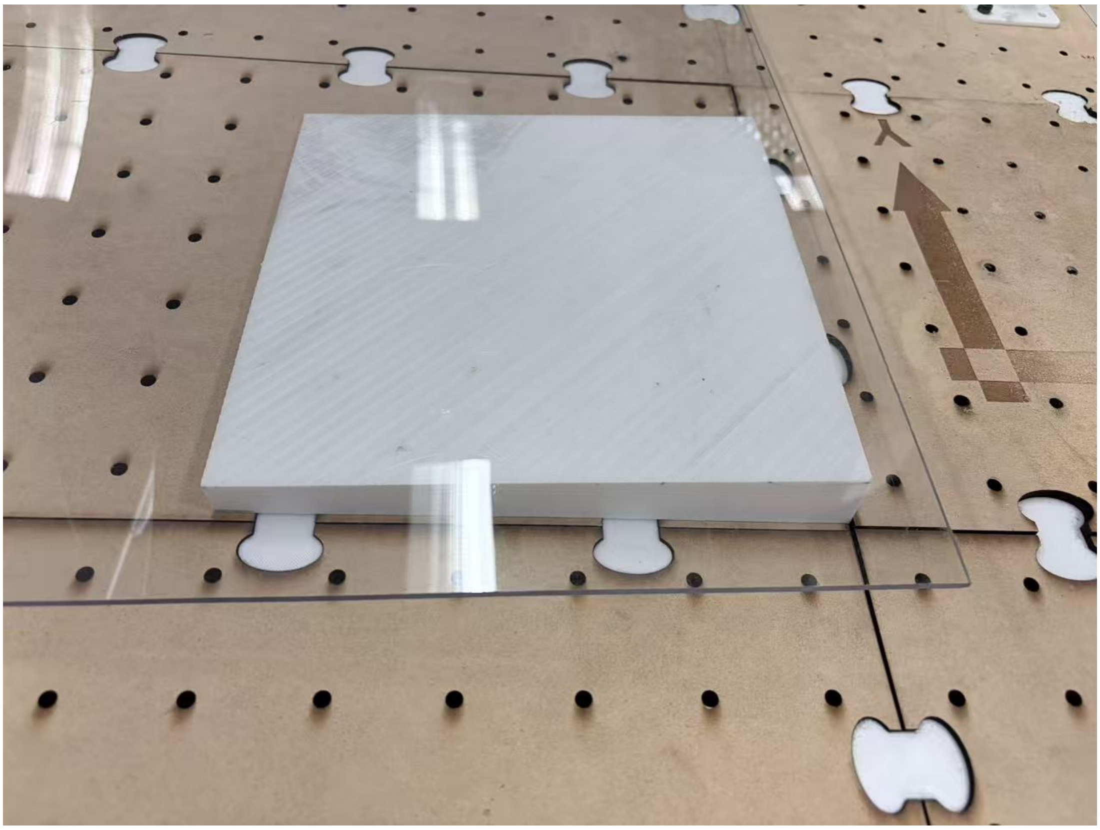


Figure 4.1: The `SURFACE_RISER` wedge (3D-printed, white) mounted under the acrylic test surface, on the optical breadboard used for Experiments 1 and 2. The arrow marks the gripper's local y axis (the gripping direction), along which the wedge slope is aligned.

[TODO: Experiment 3 is a placeholder: the protocol above is specified (Bill 0009, pending enactment) and the wedge fixture is built (Figure 4.1), but the pilot has not yet been run. Insert the outcome grid (variant $\times$ attempt, success/no-grasp/abort), the as-built incline angle if it differs from the CAD nominal 4.15°, and a short results paragraph once the session is

complete. If the pilot is not run before submission, delete this section and retain the inclined-surface demonstration as future work in Chapter 7.]

5

Results

This chapter reports the empirical findings across all six gripper variants (`finray_18`, `finray_18_model2`, `finray_26`, `finray_26_5deg`, `parallel_jaw_stock`, `parallel_jaw_TPU`) organised around the primary measurement (contact force F_z) at swept command depths and $\pm 2.5°$ tilts about the x and y axes. All values were F_z-bias-corrected using the per-session Franka F_{ext} zero-load reading and pooled across trials with matching command depths and tilts unless otherwise noted. Interpretation is deferred to Chapter 6.

5.1 Print-to-Print Manufacturing Variability

Figure 5.1 compares two independent prints of the same `finray_18` CAD geometry directly on the calibrated depth axis. Each zeroed at its own first majority-success cell (Section 3.7.1), the two prints differed by a +4.6 N mean force offset (second print stiffer) with RMSE 5.4 N over the 10 mm window; removing the constant offset left a residual RMSE of 2.9 N. This $\approx$ 5 N offset was adopted as the canonical same-session print-to-print variability, approximately 15% of the knee peak force.

Cross-session drift on the *same* print, measured by comparing `finray_18` data collected on separate days approximately three weeks apart, was approximately 7–10 N at knee and knee-adjacent depths. Cross-session drift on the same specimen was therefore larger than same-session print-to-print variability.

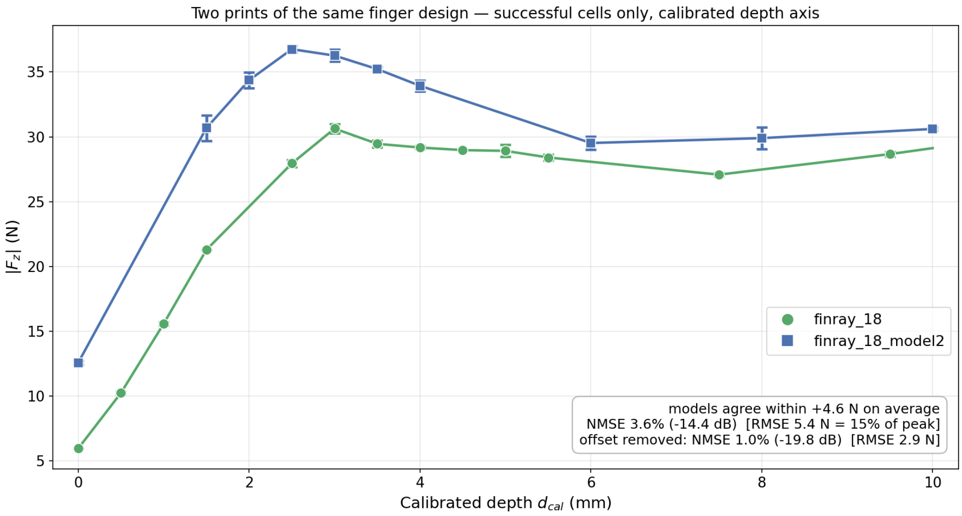


Figure 5.1: Contact force versus calibrated depth for two independently printed copies of `finray_18`, zero tilt, successful cells only. Error bars: one standard deviation across repeats; where no bar is visible, the spread is smaller than the marker.

The two prints' primary force-depth response was otherwise nearly identical (pairwise shape NMSE 0.41%, Table 5.2), and even their secondary x-axis bias shared the same depth-gated envelope as the rest of this thesis's bias measurements (Section 5.6): largest near the boundary and relaxing toward zero at depth. The two prints differed only in where that shared envelope sat relative to zero. `finray_18`'s bias crossed zero (negative at the boundary, briefly positive through the knee, negative again at depth), while `finray_18_model2`'s stayed negative throughout the full measured range; the gap between the two (2–3.5 N near the boundary, shrinking to < 1 N by $d = +10$ mm) was itself depth-gated. The attribution of this offset is deferred to Section 6.1.

5.2 Working Range at Zero Tilt

On the calibrated depth axis (Section 3.7.1), at 0° tilt the parallel-jaw variants exhibited closed successful-grasp windows of 1.5 mm (`parallel_jaw_stock`) and 2.5 mm (`parallel_jaw_TPU`), while all four finray variants spanned the full ≥ 10 mm analysis window (Figure 5.2); their ranges were right-censored at the window end, and the underlying record extends beyond it.

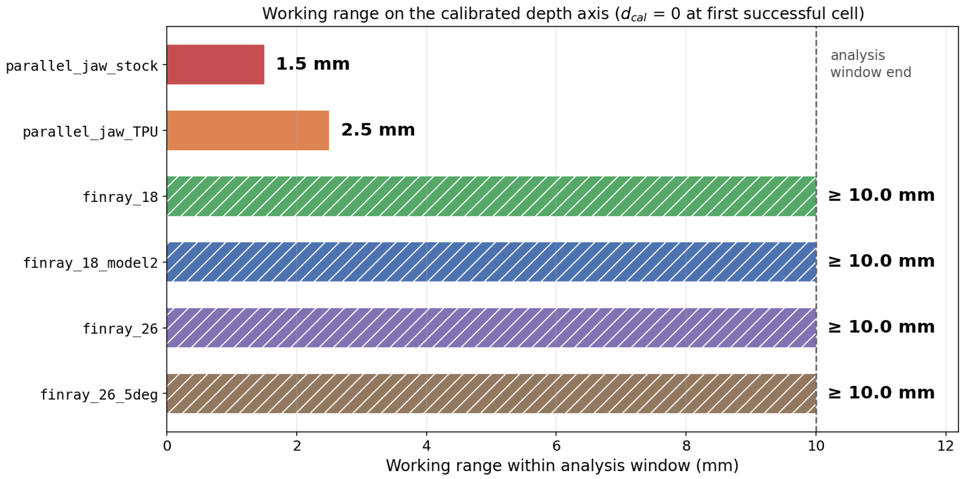


Figure 5.2: Successful-grasp working range on the calibrated depth axis, zero tilt. Hatched bars are right-censored at the 10 mm analysis-window end (Section 3.7.1).

5.3 Working Range Under Tilt

Extending the comparison to every tested orientation (five for the finrays, four for the parallel jaws, whose $\theta_x = -2.5°$ condition is covered by mirror symmetry rather than separate cells; Figure 5.3) narrowed the parallel-jaw windows further, to 0.5–2.5 mm, with one stock-jaw tilt condition reduced to a single successful depth (0 mm span) and two others producing no successful grasp at any depth. Every finray variant retained its full ≥ 10 mm window under every orientation tested.

The parallel jaws auto-failed on the Franka excess-force safety monitor (triggered at -60 N in F_z) at depths only 1–1.5 mm beyond their successful-grasp window; the finrays never triggered the safety monitor within the tested -2 to $+16$ mm range on any variant.

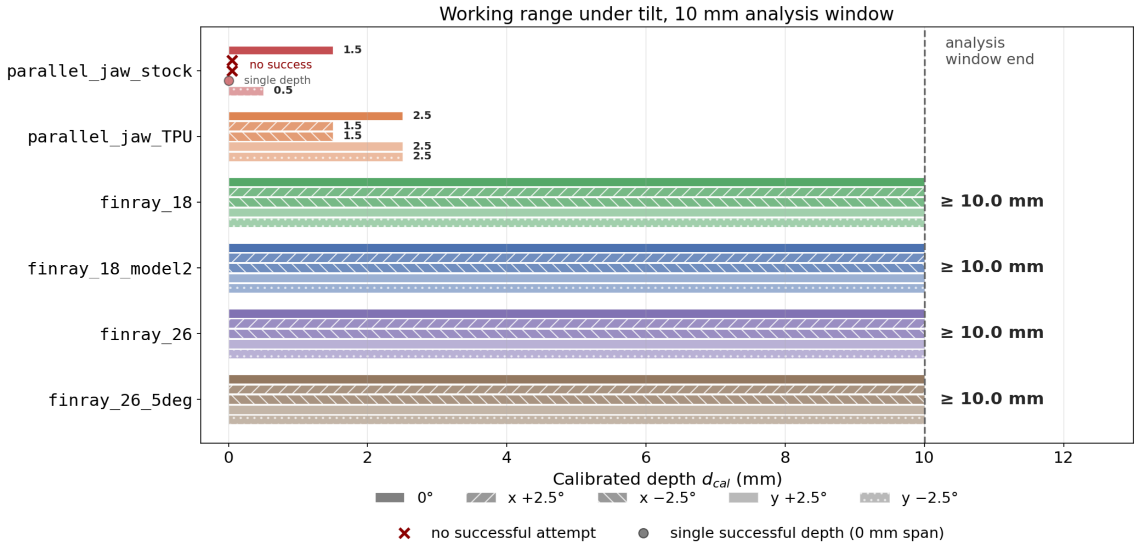


Figure 5.3: Successful-grasp working range on the calibrated depth axis, all tested orientations per variant (five for the finrays, four for the parallel jaws). ×: no successful attempt at any depth. Hatched bars are right-censored at the analysis-window end.

5.4 Tilt-Aligned Response Comparison

Plotting each finray variant's five tilt-condition curves together on the calibrated depth axis (Figure 5.4) shows how tightly the orientations track one another within a variant. The mean spread across the five orientations, averaged over the calibrated window, was 1.3–2.4 N depending on variant, small relative to the 31–53 N knee peaks reported in Table 5.1, and consistent with the depth-gated tilt robustness reported in Section 5.6: the curves separated near the boundary and converged in the plateau.

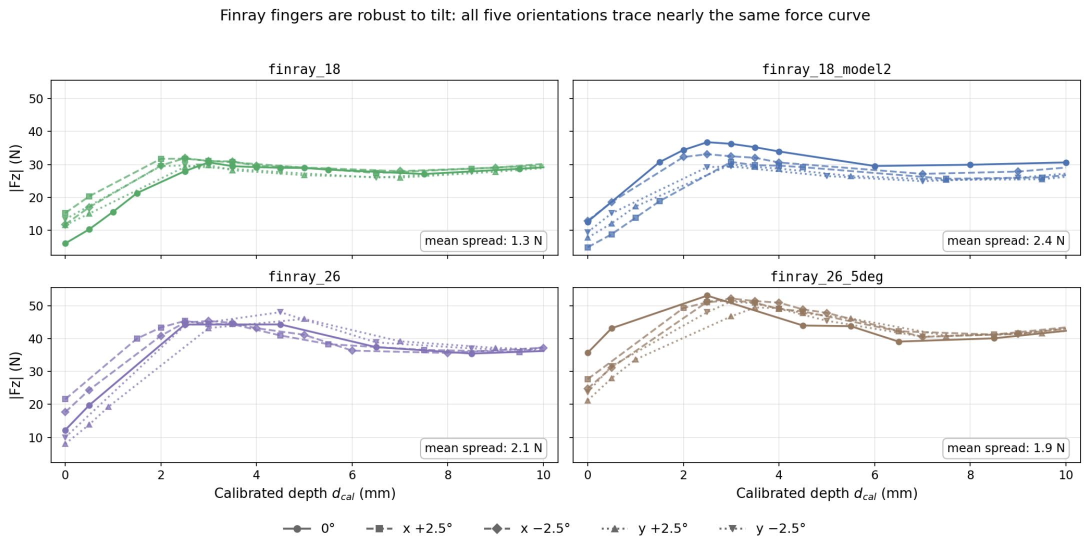


Figure 5.4: Contact force versus calibrated depth for each finray variant, all five tested orientations overlaid (0°, $x \pm 2.5°$, $y \pm 2.5°$). Within each panel the five curves nearly coincide.

5.5 One Shared Curve Shape Across the Finray Family

Under zero tilt, all four finray variants exhibited a stereotyped *bell-plus-valley* force-depth response with four regions: (i) a boundary regime where contact force rose steeply with press depth, (ii) a knee peak at calibrated depths of +2.5 to +4.5 mm, (iii) a valley trough near +6 to +8.5 mm, and (iv) a recovery regime toward the window end. Table 5.1 summarises the key features; the four features lay within $\approx \pm 1.5$ mm of the same depth locations across all four variants, and the absolute force magnitudes scaled approximately with the frontal contact area of the finger.

Normalising each variant's curve by its own peak force and averaging over its five tested orientations makes this shared shape directly visible (Figure 5.5): all four finray variants collapse onto a single rise–peak–valley–recovery curve over the full $[0, 10]$ mm calibrated window. Table 5.2 quantifies the pairwise agreement between variants as a normalised mean-squared error (NMSE) and, in decibels, $10 \log_{10}(\text{NMSE})$; the mean pairwise NMSE was 0.58% (−22.4 dB), with the largest disagreement between any

Table 5.1: Key features of the 0° force-depth curve for each finray variant, on the calibrated depth axis (Section 3.7.1). Peak $|F_z|$ is the maximum observed contact force in the knee region; valley $|F_z|$ is the local minimum between the knee peak and the deep-regime recovery. The `finray_26` knee is a flat plateau spanning +2.5 to +4.5 mm; its listed peak depth is the maximum-force cell.

Variant	Knee peak $\lvert F_z \rvert$	Peak depth	Valley $\lvert F_z \rvert$	Valley depth
`finray_18`	30.6 N	+3.0 mm	27.1 N	+7.5 mm
`finray_18_model2`	36.8 N	+2.5 mm	29.5 N	+6.0 mm
`finray_26`	44.3 N	+4.5 mm	35.5 N	+8.5 mm
`finray_26_5deg`	53.1 N	+2.5 mm	39.1 N	+6.5 mm

two variants at 0.79% (−21.0 dB).

Table 5.2: Pairwise shape agreement between finray variants over the full $[0, 10]$ mm calibrated window: NMSE (top), NMSE in decibels (middle), and RMSE as a percentage of peak (bottom, in brackets).

	`finray_18`	`finray_18_model2`	`finray_26`	`finray_26_5deg`
`finray_18`	—	0.41% (−23.9 dB) [5.5%]	0.69% (−21.6 dB) [7.2%]	0.79% (−21.0 dB) [7.8%]
`finray_18_model2`		—	0.13% (−28.8 dB) [3.1%]	0.70% (−21.6 dB) [7.1%]
`finray_26`			—	0.75% (−21.2 dB) [7.4%]
`finray_26_5deg`				—

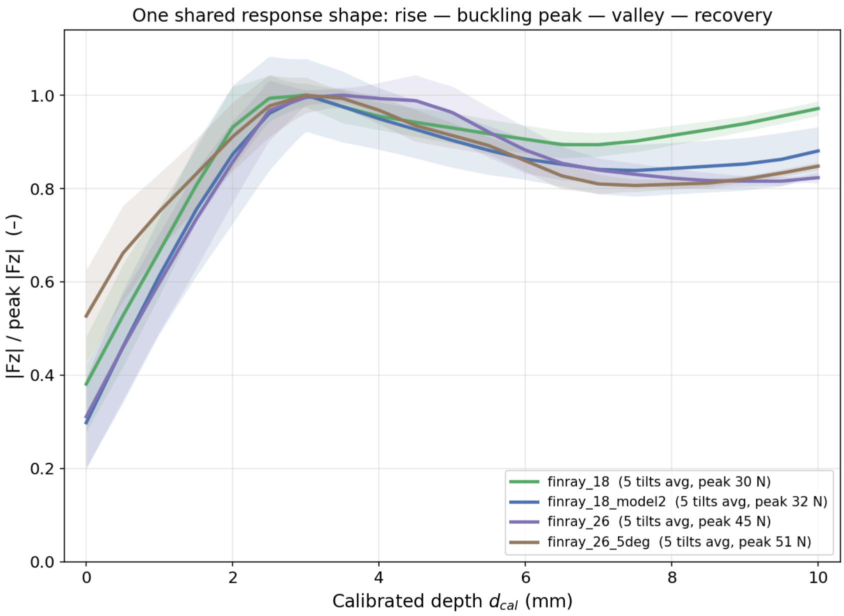


Figure 5.5: Peak-normalised $|F_z|$ versus calibrated depth d_{cal}, averaged over all five tested orientations per variant (shaded band: across-orientation spread). All four finray variants collapse onto one shared response shape.

5.6 Bipolar Tilt Bias on the Finray Family

A $\pm 2.5°$ tilt produced a signed bias $b(d,a) = |F_z^{+2.5°}(d,a)| - |F_z^{-2.5°}(d,a)|$. Depths in this section and the next are quoted on the raw command-depth axis d, the native axis of the per-cell bias record, rather than the calibrated axis d_{cal} used elsewhere in this chapter. The bias's depth trajectory itself followed a bipolar shape: a strong signed peak at boundary depths, a zero-crossing near the knee, a sign-reversed plateau through the valley, and depth-gated attenuation toward zero at deep press. The clearest example was the y-axis bias on `finray_26_5deg`, which had full coverage across 14 sign-flip pairs from -2 to $+16$ mm:

- Boundary peak: $b = +5.40 \pm 0.12$ N at $d = -1.5$ mm.
- Zero crossing: between $d = -0.5$ mm ($b = +2.00$ N) and $d = +1.5$ mm ($b = -1.35$ N).
- Knee plateau: $b = -1.57 \pm 0.21$ N at $d = +2.0$ mm, persisting to $d = +3.0$ mm.

- Depth-gated recovery: $|b| < 0.9\,\mathrm{N}$ at all depths beyond $d = +4\,\mathrm{mm}$.

The x-axis bias on `finray_26_5deg` had a similar bipolar shape but with opposite sign at the boundary ($b_x = -4.17 \pm 0.32\,\mathrm{N}$ at $d = -1.5\,\mathrm{mm}$).

5.7 Cross-Axis Sign Reversal at the Boundary

At boundary depths, the x-axis and y-axis biases had *opposite signs* on `finray_26_5deg` (Table 5.3).

Table 5.3: Bias on `finray_26_5deg` at boundary depths. x-axis and y-axis biases had opposite signs (sign product -1). Values are $|F_z^{+2.5°}| - |F_z^{-2.5°}|$ in newtons.

Depth	x-bias	y-bias	Sign product
$-1.5\,\mathrm{mm}$	-4.17 ± 0.32	$+5.40 \pm 0.12$	-1
$-1.0\,\mathrm{mm}$	-3.94 ± 0.18	$+4.27 \pm 0.07$	-1
$-0.5\,\mathrm{mm}$	-3.51 ± 0.39	$+2.00 \pm 0.24$	-1

The same sign-reversal pattern was present on `finray_18` at $d = -1.0\,\mathrm{mm}$ (x-bias $= -2.44 \pm 0.17\,\mathrm{N}$; y-bias $= +4.49 \pm 0.47\,\mathrm{N}$), demonstrating that the effect was not specific to the 5° mount condition.

In absolute terms, the boundary y-axis asymmetry corresponds to a 2.36× force ratio between the two tilt signs: on `finray_26` at $d = -1.0\,\mathrm{mm}$, the mean stationary force was 8.00 N under +2.5° versus 3.39 N under −2.5° (Table 4.2); at $d = +10.0\,\mathrm{mm}$ the same pair differed by under 1% (36.26 versus 35.95 N). This ratio is the source of the handedness figure quoted in the abstract, and its collapse from 2.36× to 1.01× between boundary and plateau is the per-cell view of the depth gating reported in Section 5.6.

5.8 Mount-Tilt Alignment: 5° Mount as a Depth Shift

After a two-parameter best-fit alignment (single depth shift plus single force offset), the 5°-mounted variant collapsed onto the flat-mounted variant with residual RMSE below 2 N on both tilt axes independently (Table 5.4).

Table 5.4: Best-fit alignment of `finray_26_5deg` onto `finray_26`. The two axes yielded independent estimates of the depth shift within 0.02 mm of each other.

Axis	Depth shift	Force offset	Alignment RMSE
x	−1.14 mm	+2.7 N	1.75 N
y	−1.12 mm	+3.1 N	1.58 N
mean	**−1.13 mm**	+2.9 N	1.66 N

6

Discussion

This chapter interprets the results in the order Chapter 5 reported them: the error structure of the measurements, the working range at zero tilt, the effect of tilting, the similarity of the finray responses across orientation and depth, and finally the shared force-depth profile and the buckling mechanics behind it. The limitations bounding these claims close the chapter (Section 6.6).

6.1 Measurement and Manufacturing Error

The print-to-print comparison of Section 5.1 doubles as the error budget for the whole dataset, and its central finding is a noise hierarchy. Same-session print-to-print variability ($\approx$ 4.6 N mean offset, RMSE 5.4 N) proved *smaller* than cross-session drift on the same print (7–10 N at knee depths). Time, not manufacturing, is the dominant nuisance variable: candidate contributors include cloth conditioning, TPU viscoelastic recovery, and drift in the FCI external-wrench estimate, which is a model-based estimate rather than a load-cell reading [46]. Two rules follow. Any cross-variant force claim with an effect size below the $\approx$ 5 N print-to-print offset is unlikely to reproduce across physical instances of the same CAD, and should be treated as tentative until replicated across prints. And any campaign that pools data across days must either report cross-day drift as a separate variance component or restrict its headline claims to same-session pairs; in this thesis, all bias values are same-session unless labelled otherwise, and cross-day comparisons are used only for shape consistency. Table 6.1 collects the error terms on a common scale.

Table 6.1: Error and variability terms of the campaign, ordered by magnitude. Percentages are relative to a nominal 30 N knee peak. The shape terms are computed on peak-normalised curves and so are insensitive to the force offsets above them.

Term	Magnitude	Relative	Source
Cross-session drift, same print	7–10 N	23–33%	Section 5.1
Print-to-print offset, same session	4.6 N	15%	Section 5.1
residual after offset removal	2.9 N	10%	Section 5.1
Boundary tilt bias, peak	5.4 N	18%	Section 5.6
Plateau tilt bias	< 0.9 N	$< 3\%$	Section 5.6
Across-orientation spread, calibrated	1.3–2.4 N	4–8%	Section 5.4
Within-cell retest trigger	1 N	3%	Section 4.2
Within-trial sample noise	~ 0.1 N	$< 1\%$	Section 3.7
Print-to-print shape (NMSE)	0.41%	−23.9 dB	Table 5.2
Cross-variant shape (NMSE, mean)	0.58%	−22.4 dB	Table 5.2
Cross-variant shape (NMSE, worst)	0.79%	−21.0 dB	Table 5.2

The table makes the central asymmetry of the dataset visible at a glance. Every force-magnitude term sits between 1 and 10 N, one to two orders of magnitude above the shape terms, which are below 1% normalised error. Absolute forces in this campaign are therefore uncertain at the several-newton level, while the *shape* of the force-depth response is reproducible to a fraction of a percent. This is why the thesis leans on shape agreement and depth locations for its headline claims and treats absolute force comparisons as session-bounded.

The x-axis bias sign difference between the two prints, deferred here from Section 5.1, fits inside this error budget rather than outside it. Both prints' bias curves share the same depth-gated envelope, strongest at the boundary and fading with depth, and differ only in a shrinking, print-dependent offset (2–3.5 N near the boundary, < 1 N by $d = +10$ mm) that happens to carry `finray_18` across zero while finray_18_model2 remains on one side throughout. The offset lives entirely on the secondary (off-axis) force component and leaves the *shape* of the dominant F_z response essentially unchanged (NMSE 0.41%); the +4.6 N magnitude offset is a separate, primary-axis quantity, already counted in the noise hierarchy above. Two candidate mechanisms are consistent with the data: small print-instance differences in rib-wall thickness or infill density manifesting as a direction-dependent buckling offset, or a small, direction-dependent bias in the F_z-dominant external-wrench estimate (Section 3.2.1), which would naturally be largest where the signal is weakest (the boundary) and vanish where it is strongest (the plateau). Distinguishing them requires post-hoc CT scanning of the two prints or the independent force-sensor cross-check identified as future work (Chapter 7); until then, the finding is a bound on secondary-axis force uncertainty, not evidence that the two prints respond differently to tilt.

The measurement design also shapes what counts as valid data. The trial-outcome taxonomy of success, no-grasp, and failed trial (Section 4.5) is deliberately three-valued because the categories map to physically distinct regimes. Treating no-grasp trials as failures would delete clean force measurements from the boundary regime, truncating the observed curves and masking the boundary-tilt biases; treating failed trials as no-grasps would pollute the averages with readings from incomplete stationary windows (near-zero from interrupted trials, saturated from safety-reflex aborts). The three-way taxonomy is the minimum classification that separates "the trial produced valid measurement" from "the grasp was successful" from "the trial was invalid."

6.2 Working Range Without Tilt

At zero tilt the two gripper classes occupy different regimes of the same axis: closed windows of 1.5 and 2.5 mm for the rigid jaws against censored windows of at least 10 mm for every finray (Section 5.2). The contrast in failure behaviour matters as much as the widths. The parallel jaws auto-failed on the −60 N excess-force reflex within 1–1.5 mm of their window edge, while the finrays never triggered the reflex anywhere in the tested −2 to +16 mm range. Compliance therefore flattens the map from position error to force spike, which for collaborative deployment is arguably more valuable than the success rate itself.

The deployment reading is direct. Finray-based systems can run open-loop, with no depth-clipping controller and no safety-limit tuning. Parallel-jaw systems require an active depth governor with command precision well inside their 0.5–2.5 mm tilt-dependent window, since the jaws auto-failed within 1–1.5 mm beyond the window edge, leaving little margin for command error. Any cloth-handling system whose depth uncertainty exceeds roughly half the rigid-jaw window should default to a finray-class gripper. The no-grasp regime at the window's shallow edge is also where a grasp controller could in principle intervene productively: the pre-lift press force is measured and the depth is known, so a marginal engagement could be flagged before the lift begins, and the controller could deepen the press or abort rather than drop the cloth outside the pick zone.

6.3 Effect of Tilting

Tilting the arm degraded the two classes asymmetrically: the rigid windows shrank to 0.5–2.5 mm, with one stock-jaw condition reduced to a single successful depth and two producing no success at all, while every finray variant kept its full window at every orientation (Section 5.3). Within the finrays, the tilt effect is depth-gated. The bipolar bias trajectory of Section 5.6 (a strong signed asymmetry at boundary depths that attenuates to under 0.9 N beyond $d = +4$ mm, raw command depth being the native axis of the per-cell bias record) tracks the contact-geometry transition documented in Figure 3.4. At boundary depths only the blade tips touch the surface: the contact patch is small, and a 2.5° orientation change redistributes which tip carries the load, so the asymmetric blade profile dominates the force response. Past the buckling knee, the blades wrap the surface and the contact becomes an extended line along each blade; the same orientation change then only slightly redistributes load along an already-engaged contact. The depth at which the bias attenuates (raw $d = +4$ mm, about +4.5 mm calibrated for this series) sits just past the knee-peak region, consistent with wrap-around engagement, rather than press depth per se, being the quantity that confers tilt robustness. This is the mechanism behind the depth-gated tilt robustness identified as the principal empirical contribution of this thesis (Chapter 7): compliance does not remove tilt sensitivity, it relocates it to the contact-boundary regime that a sufficiently deep press avoids.

What, mechanically, is the residual bias? Comparing the raw-axis bias curves with the per-series calibrated view of Section 5.4 implies that the raw-axis asymmetry is, to first order, an orientation-dependent shift in engagement onset seen through the local slope of the bell curve, rather than a change in the force law itself. This inference is not circular: the per-series zero is defined by grasp success, not by any feature of the force curve, so the collapse of the calibrated curves is evidence that the success onset and the force-curve position shift together under tilt. The reading is quantitatively self-consistent (dividing the $+5.40\,\mathrm{N}$ boundary bias by the boundary-region slope of ≈ 5–$10\,\mathrm{N/mm}$ implies an onset shift of 0.5–1 mm, at or below the cell spacing) and it organises the bipolar trajectory: the bias is large where the curve is steep, reverses sign as the curve tops out approaching the knee (the measured crossing between $d = -0.5$ and $+1.5\,\mathrm{mm}$ precedes the raw-axis knee peak of $\approx +2\,\mathrm{mm}$ by about a millimetre, consistent with the knee flattening ahead of its peak cell), and attenuates at depth faster than the local slope alone would predict, indicating that the onset shift itself collapses as the blades wrap. Two honesty notes bound the claim: the 1.3–2.4 N residual spreads set the floor above which bias structure can be resolved, so only the boundary magnitude and the sign reversal are established above that floor; and the asymmetry remains operationally decisive regardless of interpretation, because a deployed system calibrates once, blind to its own tilt error, and therefore lives on the raw axis.

The sign structure of the asymmetries separates their origins. The y-axis bias is intrinsic to the finger: the gripper is mirror-symmetric about the xz plane (Section 3.2.1), so rotation about x, the symmetric axis, is sign-equivalent by construction, while rotation about y, the gripping direction, is not: a $+2.5^\circ$ rotation about the gripping axis loads the two fingers differently and produces a signed force difference. The x-axis bias should vanish by that same symmetry, so its nonzero value (of opposite sign to the y-bias, Section 5.7) indicates an additional symmetry-breaking contribution. Two hypotheses remain live. The cloth may carry an anisotropic warp–weft axis misaligned with the gripper's mirror plane, coupling through the whole-hand contact into a signed force difference. Alternatively the asymmetry may sit on the finger or fixture side after all: the tilt-geometry verification (Table 4.2) already established a sign-independent $+1.2\,\mathrm{mm}$ common-mode deflection in $+x$ under $\sim 36\,\mathrm{N}$ press, and a fixed structural lean of this kind would also produce a signed x-bias at the boundary, since a $+2.5°$ x-tilt deepens the preferred lean while a $-2.5°$ tilt opposes it. The reproducibility of the x-bias across two variants does not discriminate, because both share the same seat design, printer, and process. Rotating the cloth specimen 90° in the lab frame is the discriminating experiment: a cloth-side origin flips the x-bias sign, a finger-side origin leaves it unchanged, and the y-bias should be approximately unchanged in either case.

6.4 Similarity Across Orientation and Depth

Once each series is re-zeroed at its own first majority-success cell, the five orientation curves of every finray variant nearly coincide, with mean spreads of only 1.3–2.4 N against knee peaks of 31–53 N (Section 5.4). In other words, orientation moves the engagement onset but barely perturbs the force law that follows it. The per-series calibration of Section 3.7.1 is therefore doing real scientific work: its success in collapsing the orientations (and the two prints) is itself evidence that engagement onset is the physically meaningful origin for this contact problem.

The seat-mount comparison extends the same conclusion to a fixed, built-in tilt. Raising the seat angle from the $+2.5°$ baseline to $+5.0°$ proved equivalent to a -1.13 mm depth shift, axis-independent to within 0.02 mm (Table 5.4). Because the comparison variant already sits on the $+2.5°$ baseline seat (Section 3.1.2), the differential tilt is 2.5°; a differential tilt of a rigid finger of length L produces a projected vertical offset of $L\,[\sin(5°) - \sin(2.5°)] = 0.0436L$, and setting this equal to the observed shift gives $L_{\text{eff}} \approx 26$ mm, near the mid-length of the ~ 50 mm finger, consistent with load distributed along the finray's curved contact surface rather than concentrated at the distal tip. The axis-independence rules out any mechanism in which the pre-tilt produces a different effective load-bearing length on the two axes: the seat-angle increment acts as a strictly rigid translation of the operating point in depth, shifting the envelope by 1.13 mm for a 2.5° increment at the cost of a ~ 3 N vertical force offset, without altering the underlying force-depth sensitivity.

6.5 One Shared Profile: Peak, Valley, Recovery, and Buckling

The deepest regularity in the dataset is that all four finray variants, two thicknesses, two prints, and two seat angles, collapse onto a single peak-normalised force-depth curve (mean pairwise NMSE 0.58%, Section 5.5). The curve's shape is consistent with the finger operating in three sequential contact regimes. In the boundary regime, only the distal tip contacts the cloth and force rises roughly linearly as contact area grows. At the knee, the full blade section is engaged and the finger resists compression through its rib structure, producing the peak. Past the peak, the ribs partially buckle and the finger folds back on itself: contact area increases but effective stiffness drops, producing the valley. At recovery depths the finger has fully wrapped, and further penetration is resisted by structural compression through the finger base. That the four features (boundary, knee peak, valley, recovery) occur at the same calibrated depths ($\approx \pm 1.5\,\mathrm{mm}$) across all variants indicates the three-regime response is a geometric property of the finray shape, not sensitive to blade thickness or seat angle: force magnitudes scale with contact area, but the transition depths do not.

Read as structural mechanics, the curve is a buckling signature, and its non-monotonicity deserves explicit flagging: an operator reasoning from rigid contact expects force to grow monotonically with press depth, and here it does not. The boundary rise is elastic pre-buckling loading of the straight blades; the knee peak is the limit point at which the blades reach their tip-load (buckling) limit; the valley is the post-buckling softening branch, in which the buckled blades carry less force even as displacement continues (the load drop characteristic of an elastic structure past a limit point); and the recovery is post-buckling re-stiffening, as the inward-buckled blades come into extended contact with the surface (Figure 3.4) and further load transfers into structural compression through the finger base. This reading explains why the transition depths are geometry-set while the forces are not: a buckling limit is a property of the blade's shape and material, so geometrically similar blades place the peak and valley at the same depths while the critical load scales with blade section. This is exactly the scale-not-shape separation of Section 5.5. No published finray model covers this axial post-buckling regime (Chapter 2); the analytical model identified in future work (Chapter 7) would connect it to the continuum-mechanics literature on post-buckling contact.

The profile also carries a design implication. The valley is arguably the preferred operating setpoint rather than merely an acceptable one: it retains the full tilt insensitivity of the post-buckling regime while applying 20 to 30 percent less force than the knee peak (Table 5.1), a margin that matters for delicate or loosely woven fabrics.

6.6 Limitations

Single cloth specimen and single surface. The reported working ranges and cloth-fixture asymmetries were measured on one cotton cloth specimen on one acrylic table surface. The absolute numbers will shift with cloth thickness, weave direction, friction, and surface compliance. The cross-axis sign reversal discussed in Section 6.3 requires the 90°-rotated cloth control to separate the cloth contribution from the finger and fixture contribution.

Cross-session drift. As established in Section 6.1, cross-session drift on the same specimen is the dominant noise source in this dataset; cross-day claims are qualified throughout.

Tilt range censoring. No finray failure was observed at any tested orientation, so the finray tilt windows are censored in angle exactly as the depth windows are censored at 10 mm: the data establish a floor of 2.5° runtime tilt (with 5° of seat inclination also tolerated), not an envelope edge. The angle at which finray grasping degrades is unmeasured; Experiment 3's 4.15° incline probes only one point beyond the tested range.

Parallel-jaw x-axis coverage. The parallel-jaw variants were not swept across the x-axis tilt sign; the parallel-jaw bias values are restricted to the y-axis. Whether the x-axis bipolar bias observed on the finrays has an analogue on the rigid grippers is untested.

Boundary-depth sample sizes. Boundary-depth cells frequently have $n = 3$, and boundary cells routinely triggered the $> 1\,\mathrm{N}$ within-cell retest criterion (Section 4.2), driven by cloth-fold state variability that is not further characterised. Boundary values are honest measurements with correspondingly wide error bars.

Success rate as a binary rating. The success/no-grasp distinction is operatorassessed after the lift stage; a finer-grained grip-force-during-lift measurement would resolve marginal cases that the current binary rating collapses.

6.7 Code and Data Availability

All source code, raw trial data, ground-truth JSON files, and analysis scripts are archived in the project repository; the public repository URL and the commit hash against which this thesis was written will be stated here at final submission. **[TODO: Insert public GitHub URL and commit hash before final submission.]**

7

Conclusion and Future Work

7.1 Answers to the Research Questions

This thesis presented a controlled benchmark of six gripper variants on a single platform (Franka FR3), measuring stationary contact force and grasp outcome as a function of press depth and tilt angle over 845 analysed trials. Returning to the four research questions posed in Chapter 1:

1. *How does contact force vary with press depth for each finger design?* Every finray variant follows a single shared rise–peak–valley–recovery curve (pairwise NMSE below 1% over the full 10 mm calibrated window), whereas the rigid parallel jaws ramp steeply toward the platform's excess-force limit within roughly 2 mm of first contact.

2. *How much can the gripper be tilted before the grasp becomes unreliable?* The finrays grasp reliably across every orientation tested ($\pm 2.5°$ about both axes, plus a $5°$ seat mount) over the full ≥ 10 mm window; the rigid jaws never exceed

2.5 mm, and two stock-jaw tilt conditions produced no successful grasp at any depth.

3. *Does tilt tolerance depend on press depth?* Yes: robustness is depth-gated; near the contact boundary, tilt strongly modulates both force and success (including a 2.4× direction-dependent asymmetry about the geometrically asymmetric axis), whereas past the buckling threshold the mean across-orientation force spread shrinks to 1.3–2.4 N.

4. *What is the compliant finger's response under compression?* Blade buckling: force peaks at buckling onset (≈ 2–3 mm past first successful grasp), dips as the blades buckle, then recovers. The finray geometry (blade thickness, seat angle, print instance) sets the force *scale* (31–53 N knee peaks) but not the curve *shape*.

The depth-dependent tilt robustness identified here had not been quantitatively characterised in prior literature, to the author's knowledge, and constitutes the principal empirical contribution.

7.2 Implications

With the robot, control policy, cloth, and environment held identical, a finger swap alone buys at least a 4× wider press-depth tolerance than the best rigid configuration (≥ 10 mm versus 2.5 mm on the padded jaws, and roughly 7× versus the stock jaws), at roughly half the grip force. Under tilt the advantage widens further, to 20× or more against the stock jaws' surviving conditions; all of these ratios are lower bounds, because the finray ranges are right-censored at the analysis-window end. For the application domains that motivated this work (laundry and textile handling, hospital bedding, assistive robotics, garment manufacturing, and soft-goods packaging), this means the flat-cloth pick, the step that currently keeps such systems human-started, becomes accessible to a standard arm running an unchanged policy, before any investment in cameras, tactile sensing, or learned control. The accompanying design rule is equally simple: press the compliant finger past its buckling peak (≈ 2–3 mm beyond first contact), and the grasp becomes robust to both depth and tilt placement error.

7.3 Future Work

Independent force-sensor cross-check. All force measurements derive from the arm's joint-torque-based external-wrench estimate. Repeating a subset of the matrix with a dedicated force/torque sensor at the wrist (or an instrumented surface) would bound the estimator's pose-dependent bias directly.

Task-level evaluation. Downstream task-level evaluation (folding a cloth diagonally from a corner grasp, dropping a centre-grasped cloth onto a curved object, dragging between positions under the same policy) would test whether the contact-phase advantage measured here translates into end-to-end task success, including under learned policies.

Task-level precision comparison. Beyond success rates, a soft versus rigid comparison of task-level *precision* (placement accuracy after fold or drop) would characterise the cost side of compliance.

Dynamic and high-rate behaviour. All trials here are quasi-static. The perturbation behaviour of compliant fingers under fast actuation and high-frequency control, where the compliance that absorbs placement error may instead cause oscillation or delayed settling, is uncharacterised.

Tilt-sensitivity model. An analytical model relating finger curvature, contact compliance, and tilt-induced contact-point migration would connect the present empirical findings to the broader continuum-mechanics literature on soft contact.

Multi-fabric study. All trials used a single cloth specimen. Replicating the matrix on fabrics of varying thickness, weave, and surface treatment would establish the applicability range of the depth–tilt-robustness curves identified here.

Closed-loop control. The boundary-depth regime, where finray directional asymmetry dominates, motivates a tilt-compensating controller using inline force feedback, validated against the open-loop boundary baseline.

Sim2real validation. A simulation of the finray finger in a compliant-contact physics engine could be calibrated against the empirical force–depth–tilt surface from this thesis, providing a digital twin for the gripper variants.

A

Detailed Coverage and Supplementary Data

This appendix records the final per-variant coverage of the experimental campaign. The table below was regenerated programmatically from the on-disk trial record (`experiments/cloth_grasp/`) on 2026-08-04, counting only active cells (staged, superseded, and backup cells excluded). A *cell* is one (variant, depth, tilt) session directory; each cell contains between one and several trials.

A.1 Per-Variant Coverage Table

In addition to the conditions tabulated above, `finray_26` carries five exploratory cells at $\theta_x = +5.0°$ (depths -0.5 to $+10\,\text{mm}$, recorded 2026-06-05) and one zero-tilt boundary-depth cell recorded within the Experiment 2 campaign; these were pilot probes of the larger-tilt regime and are not part of the primary $\pm 2.5°$ analysis.

Across both experiments the active record comprises 945 trials. This figure counts every trial directory on disk, including no-grasp outcomes (retained as valid data per

Table A.1: Experiment 1 (depth sweep at zero commanded tilt): active cell counts, sampled depth range, and total trials per variant.

Variant	Cells	Depth range (mm)	Trials
`finray_18`	17	$[-2, +14]$	51
`finray_18_model2`	28	$[-1, +14]$	86
`finray_26`	18	$[-1, +22]$	50
`finray_26_5deg`	10	$[-1, +20]$	30
`parallel_jaw_TPU`	9	$[-1, +4]$	26
`parallel_jaw_stock`	12	$[-1, +4]$	29
Total	94		272

Table A.2: Experiment 2 (tilt sweeps): active cell counts per commanded tilt condition, with sampled depth range in mm in brackets, and total trials per variant. The parallel-jaw variants could only be swept over their narrow force-limited depth window, and their $\theta_x = -2.5°$ condition is covered by mirror symmetry of the jaw about the xz plane rather than by separate cells.

Variant	θ_x=+2.5°	θ_x=−2.5°	θ_y=+2.5°	θ_y=−2.5°	Trials
`finray_18`	12 $[-1, +12]$	13 $[-2, +12]$	11 $[-2, +12]$	10 $[-2, +12]$	134
`finray_18_model2`	12 $[-2, +12]$	12 $[-2, +12]$	12 $[-2, +12]$	11 $[-2, +12]$	140
`finray_26`	14 $[0, +16]$	14 $[0, +16]$	11 $[-2, +16]$	10 $[-2, +16]$	163
`finray_26_5deg`	15 $[-2, +16]$	17 $[-2, +16]$	14 $[-2, +16]$	14 $[-2, +16]$	170
`parallel_jaw_TPU`	6 $[0, +4]$	—	5 $[0, +4]$	3 $[0, +3]$	40
`parallel_jaw_stock`	4 $[+1, +2]$	—	3 $[+1, +2]$	3 $[+1, +2]$	26
Total					673

the outcome taxonomy of Section 3.5.3) and failed trials (excluded from aggregate analyses); the trial counts quoted in Chapter 5 are the subset surviving the per-analysis exclusion rules stated there.

List of Publications

No portion of this thesis has been published, presented, or submitted for publication at the time of writing. A manuscript covering the depth–tilt benchmark results (Chapters 5 and 6) is in preparation.

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